COMPLETE SYSTEM WALKTHROUGH

Chronos Transit Engine

Finds planets in starlight

Feature Guide & Screen-by-Screen Reference · October 9, 2026

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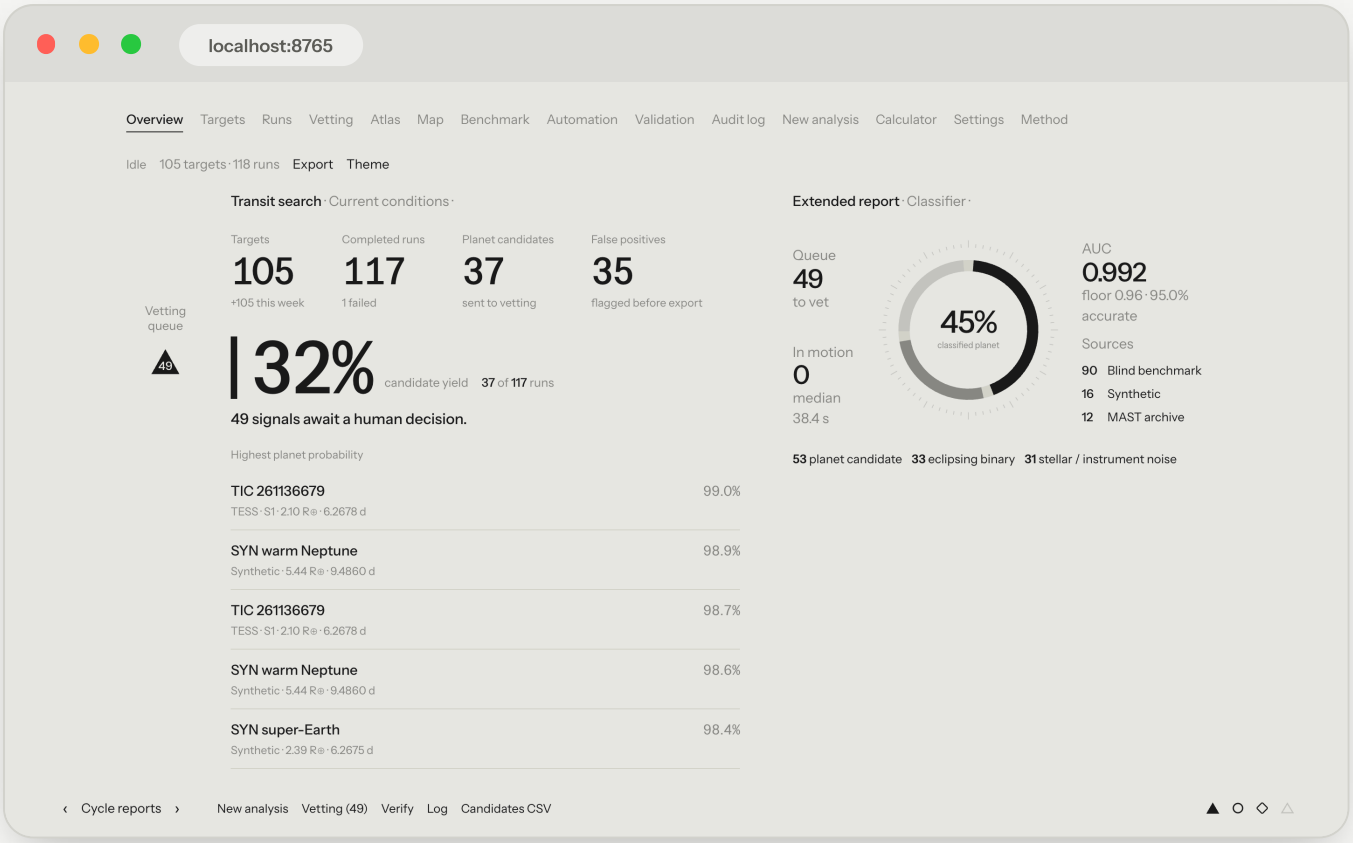
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WHAT THE SYSTEM IS

Chronos Transit Engine

Chronos Transit Engine finds planets in starlight. It takes a star's brightness over time from NASA's TESS, Kepler or K2 missions, searches it for the small, regular dips a planet makes when it crosses its star, measures the planet, and decides whether the dip is a planet, an eclipsing binary star, or noise. It is one Django project: server-rendered pages, the science pipeline, and the archive. This guide walks through every page in the order you would use them.

- **Ingest.** Any TESS, Kepler or K2 target from MAST by name, a CSV upload, or a simulated star.
- **Clean.** Median normalisation, a rolling sigma clip, and a Savitzky-Golay detrend with the transit masked.
- **Search.** Box Least Squares over 0.5 to 30 days on all CPU cores, with period-alias correction.
- **Measure.** Period, depth, duration, planet radius (with a limb-darkened fit and 1σ errors), orbit size and temperature.
- **Vet.** Odd/even depth, a secondary eclipse at any orbital phase, centroid motion, V-shape, radius and duration.
- **Classify.** A five-network ensemble of dual-branch 1D-CNNs, plus a second stage that weighs the vetting evidence.
- **Look further.** Each signal and the star's own variability are removed, then the remainder is searched again for more planets.
- **Check itself.** A blind benchmark on held-out real stars, injection and recovery, golden targets and written acceptance criteria.
- **Locate.** A zoomable map of about 4,800 real planetary systems, with distances and travel times from Earth.
- **Record.** A hash-chained JSONL audit log per run, Parquet archives, and SQLite or PostgreSQL storage.
- **Run it.** `./run.sh` starts the engine on http://127.0.0.1:8765. The first start builds a virtualenv, applies migrations and trains the classifier if no checkpoint exists.



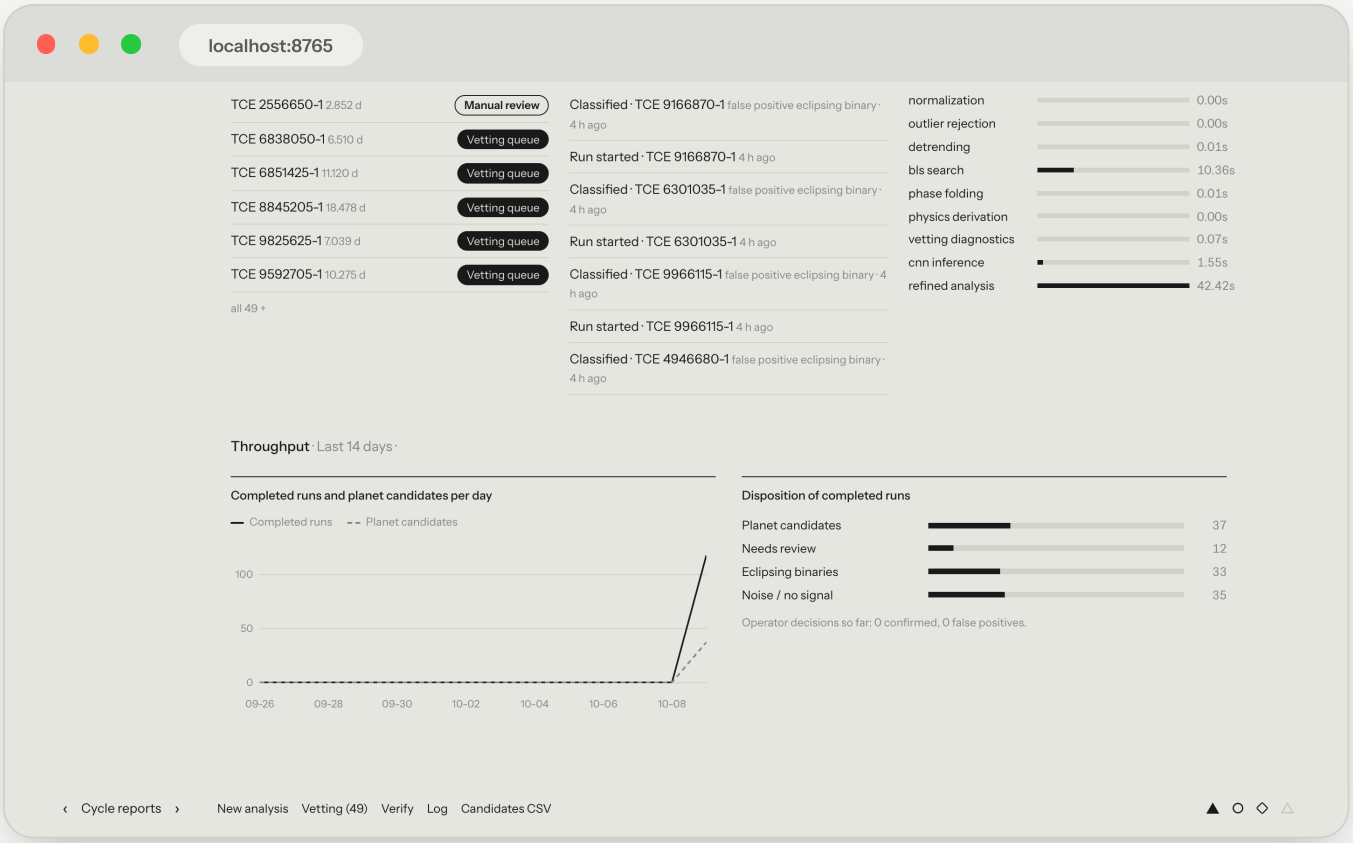
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OVERVIEW · HOME PAGE

Current conditions at a glance

The overview is the first page you see and the fastest way to read the state of the archive. The left side counts what the pipeline has done; the right side reports how the classifier is performing and where the runs came from.

- **Headline figures.** Targets, completed runs, planet candidates sent to vetting, and false positives flagged before export.
- **Candidate yield.** The share of runs that produced a planet candidate (32% here: 37 of 117), and the number of signals awaiting a human decision.
- **Highest planet probability.** The strongest signals in the archive; each row opens its run.
- **Extended report.** The vetting queue size, runs in motion, median run time, the classifier's AUC against its 0.96 floor, and the run sources (blind benchmark, synthetic, MAST archive).
- **Always on screen.** The navigation bar, the Idle / running status, Export, the light/dark Theme switch, and the dock at the bottom with shortcuts to New analysis, Vetting, Verify, Log and the candidates CSV.



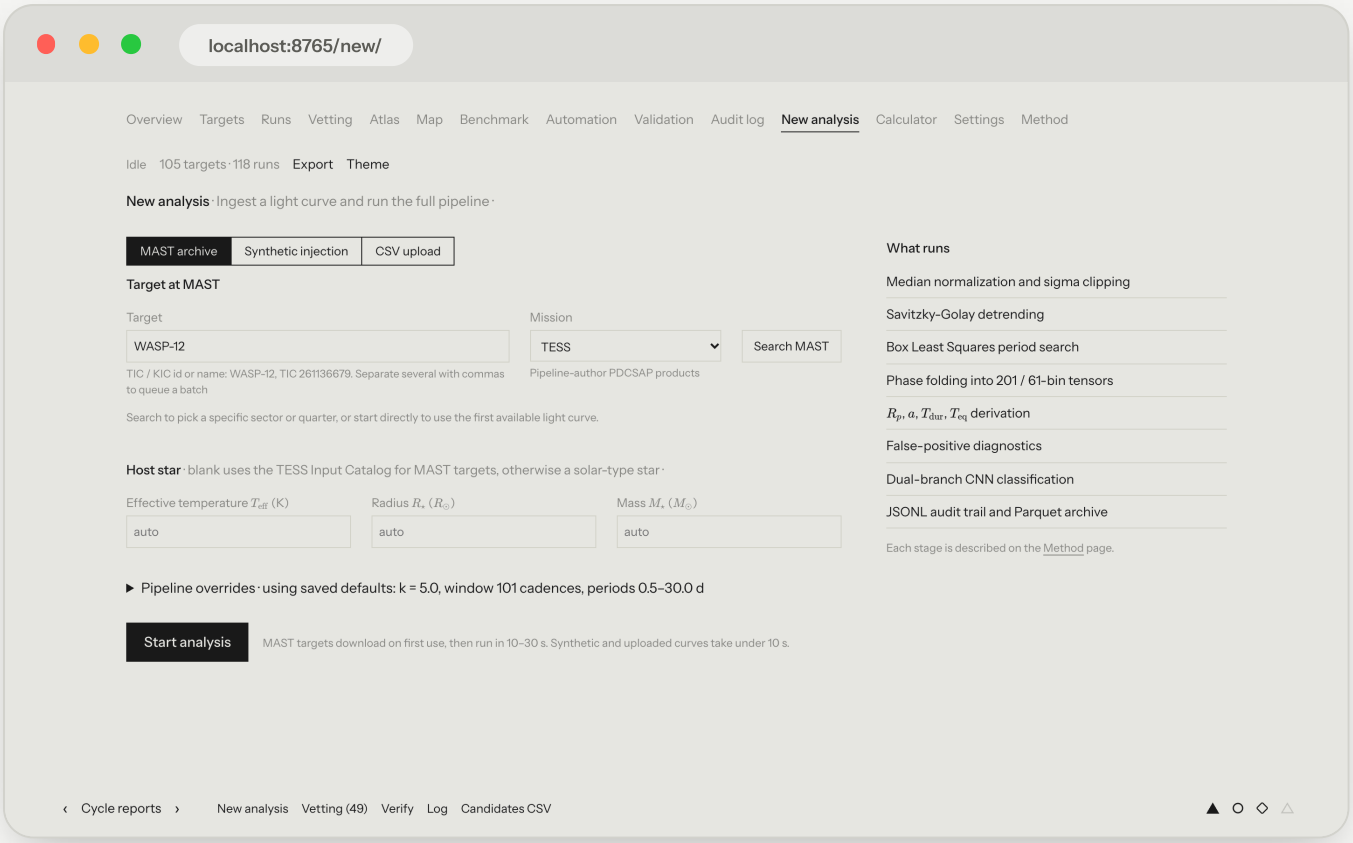
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OVERVIEW · FURTHER DOWN

Queue, activity and throughput

Scrolling the overview shows the working detail behind the headline figures: what is queued, what just happened, where the time goes, and how output has moved over the last two weeks.

- **Vetting queue.** The signals waiting for review, each tagged Vetting queue or Manual review.
- **Recent activity.** Runs starting and being classified, newest first.
- **Stage timing.** Average time per pipeline stage. The BLS period search and the refined analysis dominate; the other stages take fractions of a second.
- **Throughput.** Completed runs and planet candidates per day over the last 14 days.
- **Disposition of completed runs.** Planet candidates, needs review, eclipsing binaries and noise / no signal, plus the operator decisions made so far.



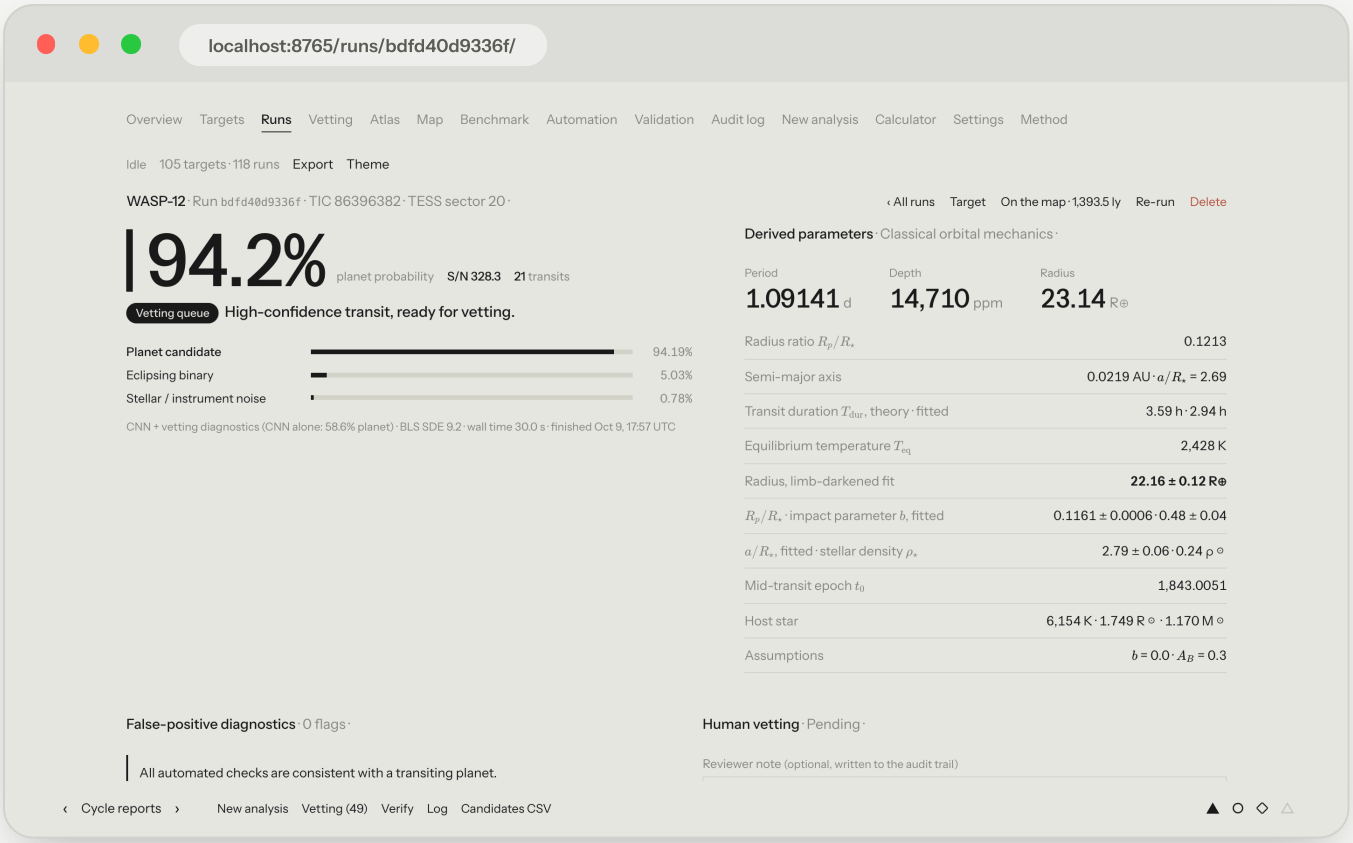
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NEW ANALYSIS · INGEST A LIGHT CURVE

Start with a star

Every run starts on this page. Choose where the photometry comes from, describe the host star if you want to override the catalogue, and start the full pipeline.

- **MAST archive.** Enter a TIC or KIC id or a name such as WASP-12, pick the mission, and optionally Search MAST to choose a specific sector or quarter. Separate several targets with commas to queue a batch.
- **Synthetic injection.** Simulate a star with a planet or binary of your choosing, useful for testing the pipeline.
- **CSV upload.** Run the pipeline on your own light curve.
- **Host star.** Effective temperature, radius and mass. Left on auto, MAST targets use the TESS Input Catalog; otherwise a solar-type star is assumed.
- **Pipeline overrides.** Per-run changes to the saved defaults (sigma-clip threshold, detrend window, period range).
- **While it runs.** The run page reports progress in plain words, shows the eleven stages and the time elapsed, and keeps a running list of what has been found so far. MAST targets download on first use, then run in 10 to 30 seconds.



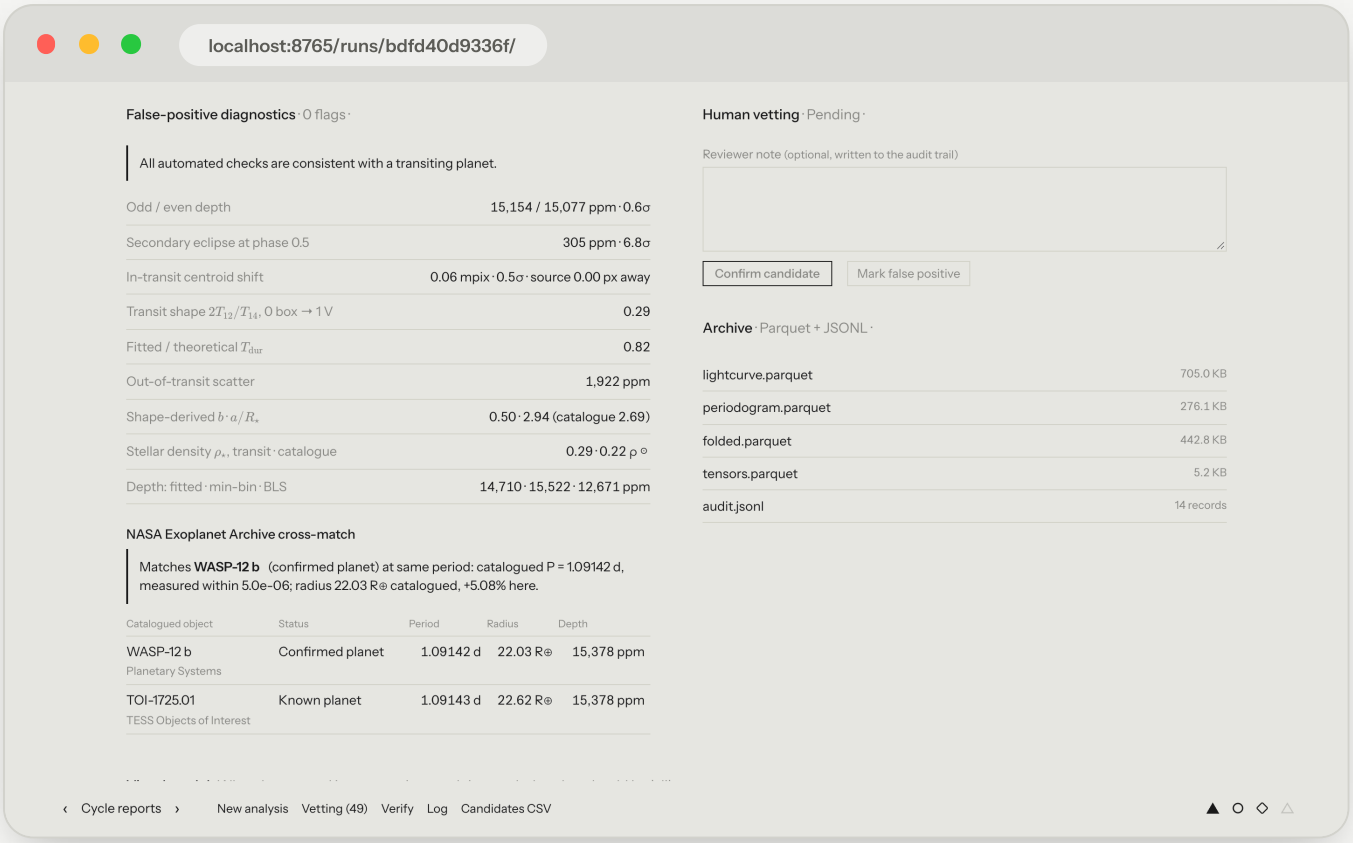
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RUN PAGE · THE ANSWER

The verdict, and the planet it measured

A finished run opens on its verdict. This example is WASP-12 from TESS sector 20: a 94.2% planet probability, a signal-to-noise of 328 from 21 transits, and the disposition Vetting queue, meaning a high-confidence transit that is ready for human review.

- **Three probabilities.** Planet candidate, eclipsing binary, and stellar or instrument noise. The line underneath shows what the CNN said alone before the vetting evidence was weighed.
- **Derived parameters.** Period 1.09141 d, depth 14,710 ppm, radius 23.14 R⊕, with the radius ratio, semi-major axis, theoretical and fitted transit duration, and equilibrium temperature.
- **Limb-darkened fit.** A second radius (22.16 ± 0.12 R⊕) with 1σ errors, plus fitted impact parameter, a/R★ and stellar density. The simple depth formula overestimates the radius by a few percent because limb darkening deepens mid-transit, so both are shown.
- **Actions.** Back to all runs, open the target, see the star on the map, re-run, or delete.



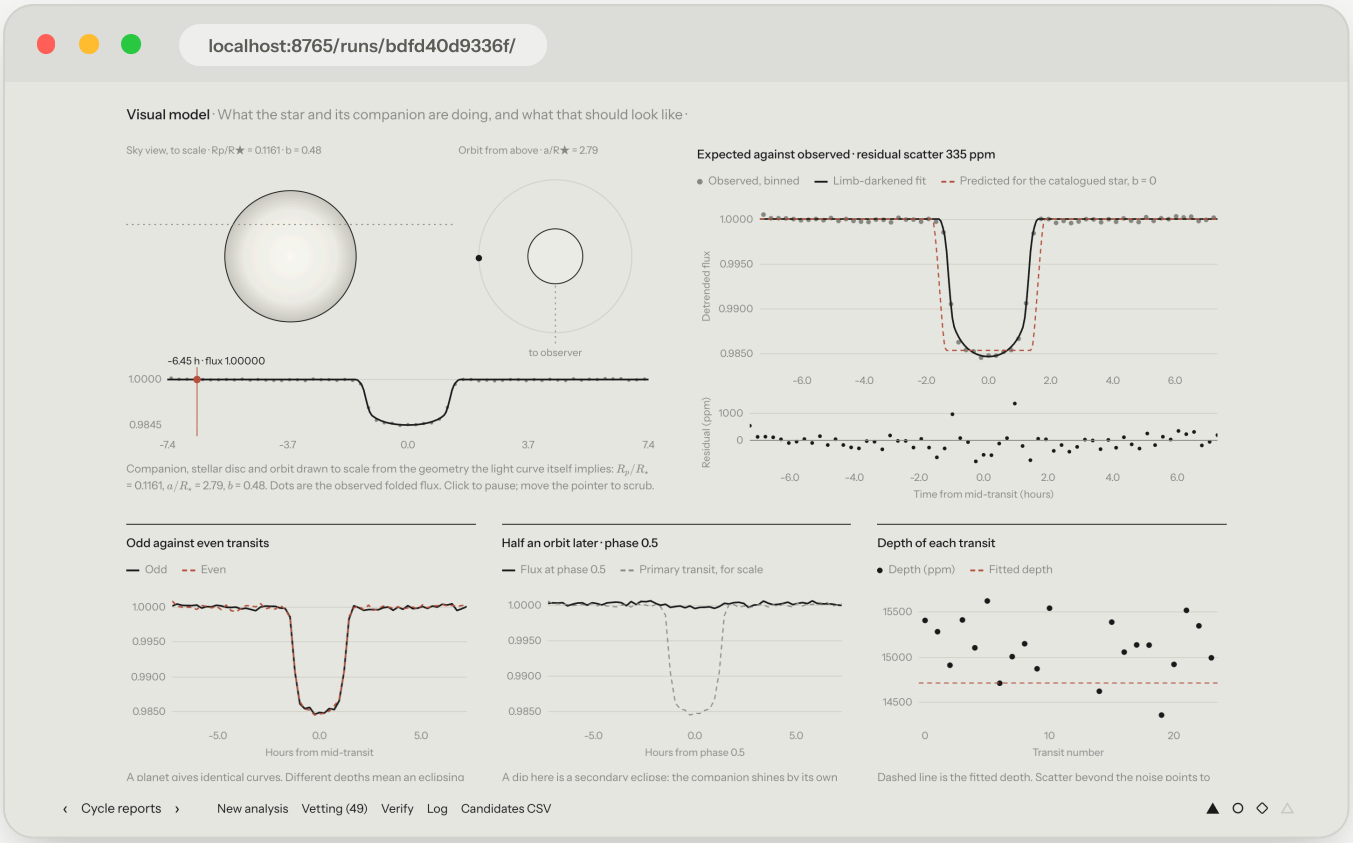
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RUN PAGE · IS IT REALLY A PLANET?

False-positive checks and the NASA cross-match

Below the verdict, the engine lists every false-positive check it made and compares its result with NASA's catalogue. Human vetting and the archived data products sit alongside.

- **False-positive diagnostics.** Odd/even depth, secondary eclipse at phase 0.5, in-transit centroid shift, transit shape, fitted against theoretical duration, out-of-transit scatter, shape-derived geometry and stellar density.
- **A real occultation is not held against a planet.** A second dip counts against a planet only when it is too bright for a planet to produce, so a hot Jupiter like WASP-12 b raises no flag.
- **NASA Exoplanet Archive cross-match.** The run is matched against confirmed planets, TOIs and KOIs. Here it matches WASP-12 b at the same period, with the radius within about 5% of the catalogued value.
- **Human vetting.** Add an optional reviewer note (written to the audit trail), then Confirm candidate or Mark false positive.
- **Archive.** Download the light curve, periodogram, folded curve and CNN tensors as Parquet, and the audit log as JSONL.



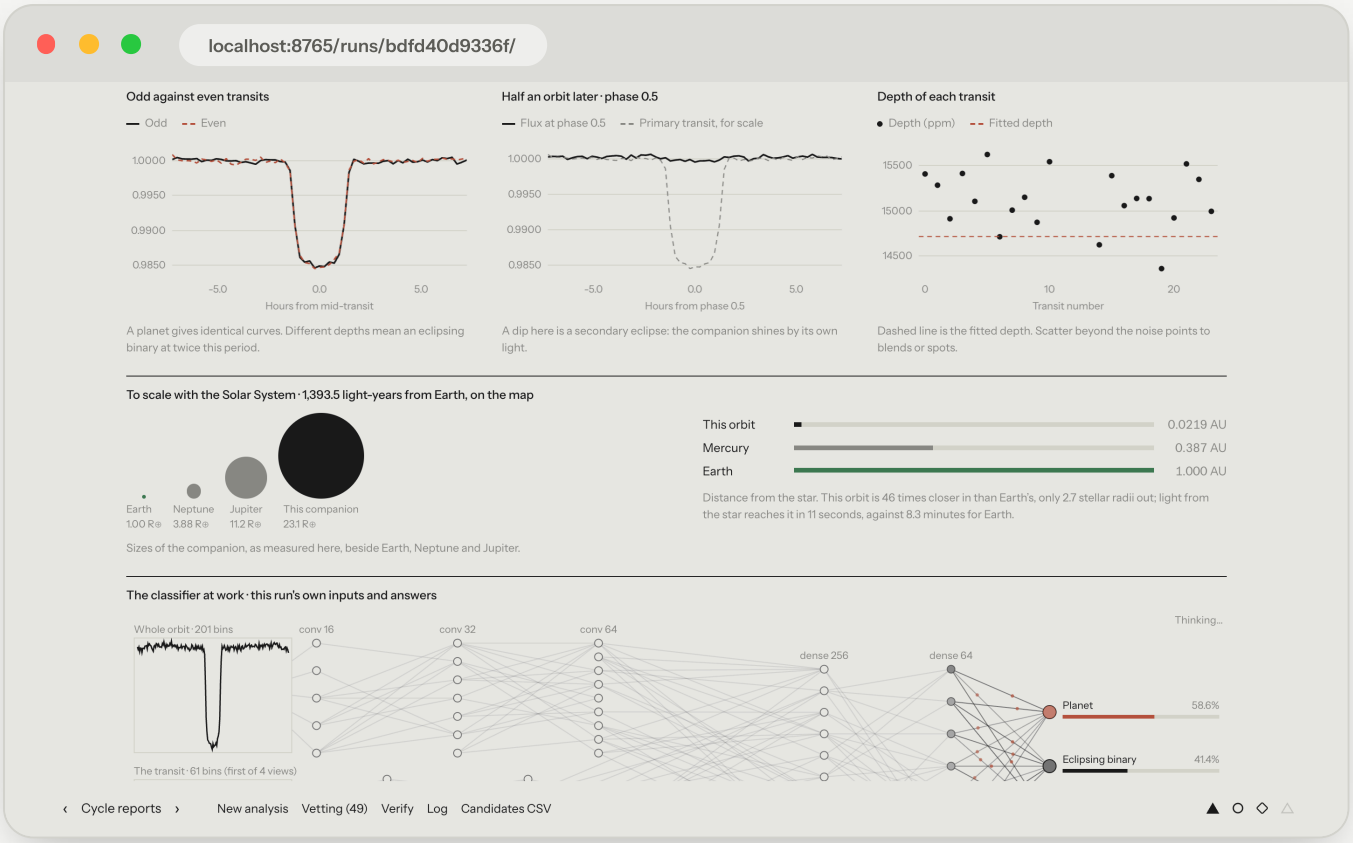
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RUN PAGE · VISUAL MODEL

A visual model, drawn to scale

The visual model turns the fitted numbers into a picture of what the star and its companion are doing, and shows how well that picture explains the data.

- **Sky view, to scale.** The companion crossing the stellar disc at the fitted radius ratio and impact parameter.
- **Orbit from above.** The orbit at the fitted a/R★, with the direction to the observer.
- **Animated light curve.** The brightness changes in step with the animation. Click to pause; move the pointer to scrub.
- **Expected against observed.** The binned observations, the limb-darkened fit, and the prediction for the catalogued star, with the residual scatter plotted below.



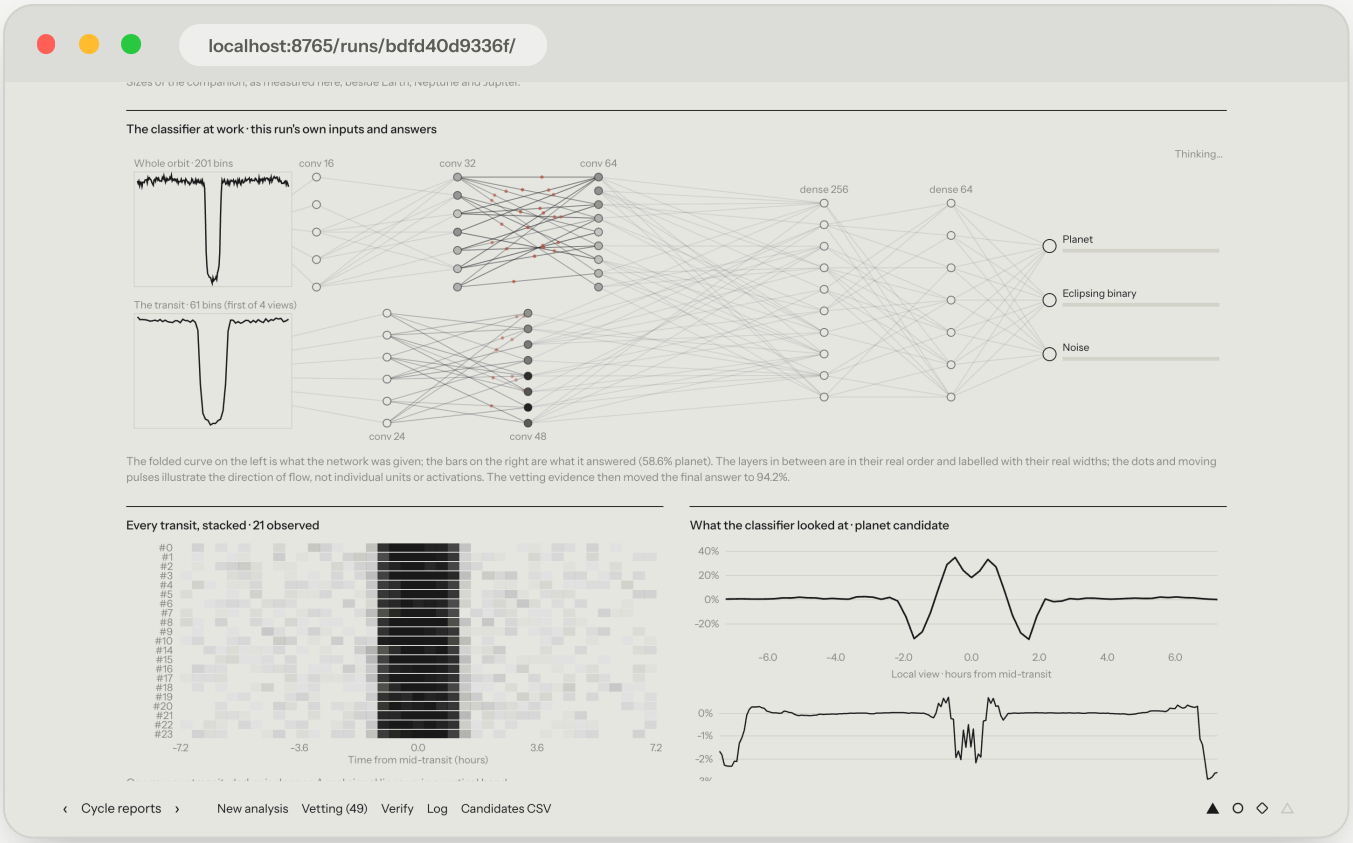
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RUN PAGE · THE EVIDENCE

The evidence, plot by plot

Three small plots carry the evidence a human vetter looks for, each with a one-line reading guide. Below them the system is put beside our own.

- **Odd against even transits.** A planet gives identical curves. Different depths mean an eclipsing binary at twice the period.
- **Half an orbit later.** A dip at phase 0.5 is a secondary eclipse: the companion shines by its own light.
- **Depth of each transit.** The dashed line is the fitted depth. Scatter beyond the noise points to blends or spots.
- **To scale with the Solar System.** The companion drawn beside Earth, Neptune and Jupiter, and its orbit beside Mercury's and Earth's. WASP-12 b orbits 46 times closer than Earth does; light from its star reaches it in 11 seconds.



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RUN PAGE · THE CLASSIFIER

The classifier at work

The classifier diagram is drawn with this run's own folded light curve as input and its own probabilities as output. While a run is in progress the same diagram sits idle until its stage arrives, then animates.

- **Two branches.** A global branch sees the whole orbit in 201 bins; a local branch sees the transit in 61 bins, in four views: all transits, odd, even, and the window half an orbit later.
- **Ensemble.** Five networks are trained, each holding out a different fifth of the real training stars.
- **Second stage.** A small ensemble combines the CNN probabilities with the vetting evidence. For WASP-12 b the CNN alone gave 58.6% planet; the vetting evidence moved the final answer to 94.2%. The final answer is what every run reports.
- **Every transit, stacked.** One row per observed transit, so a single odd event stands out.
- **What the classifier looked at.** Occlusion saliency for the local and global views: which parts of the curve moved the answer.

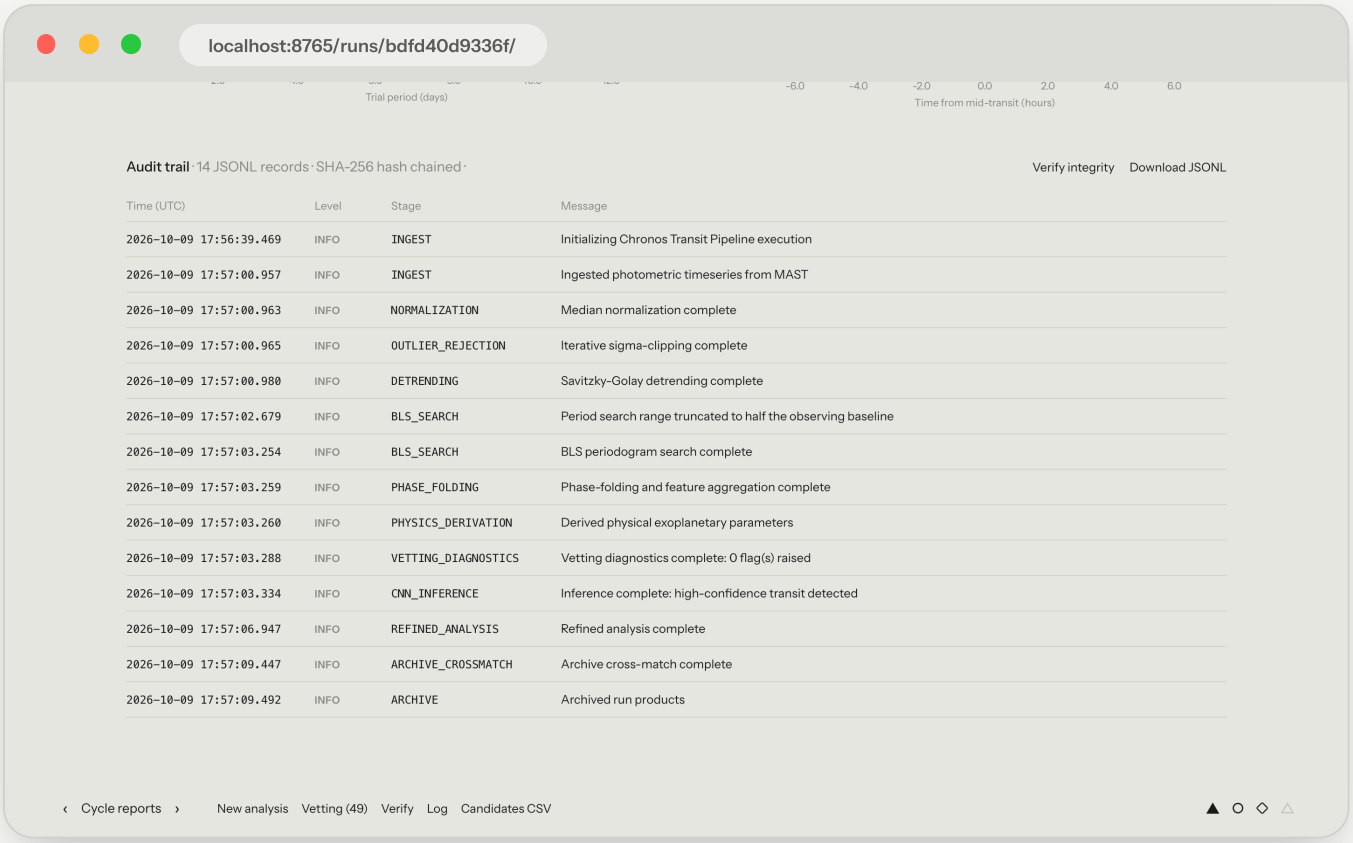


RUN PAGE · THE DATA

Light curve, periodogram and folded transit

The plots section shows the photometry the run was built from and the search that found the period. All plots can be dragged to zoom.

- **Light curve.** 16,551 cadences over 26.3 days at 2-minute cadence. Switch between Normalized + trend and Detrended. Dashed lines mark predicted transit times from the fitted ephemeris.
- **BLS periodogram.** Signal residue over 15,192 trial periods, with the best period labelled at the peak.
- **Phase-folded transit.** Every transit folded onto one curve, shown as the exact tensors the CNN receives: Local (61 bins) or Global (201 bins).
- **Period aliases.** The best peak and its simple multiples are re-examined on a fine grid, and the period is doubled or tripled when only every second or third epoch shows a transit.



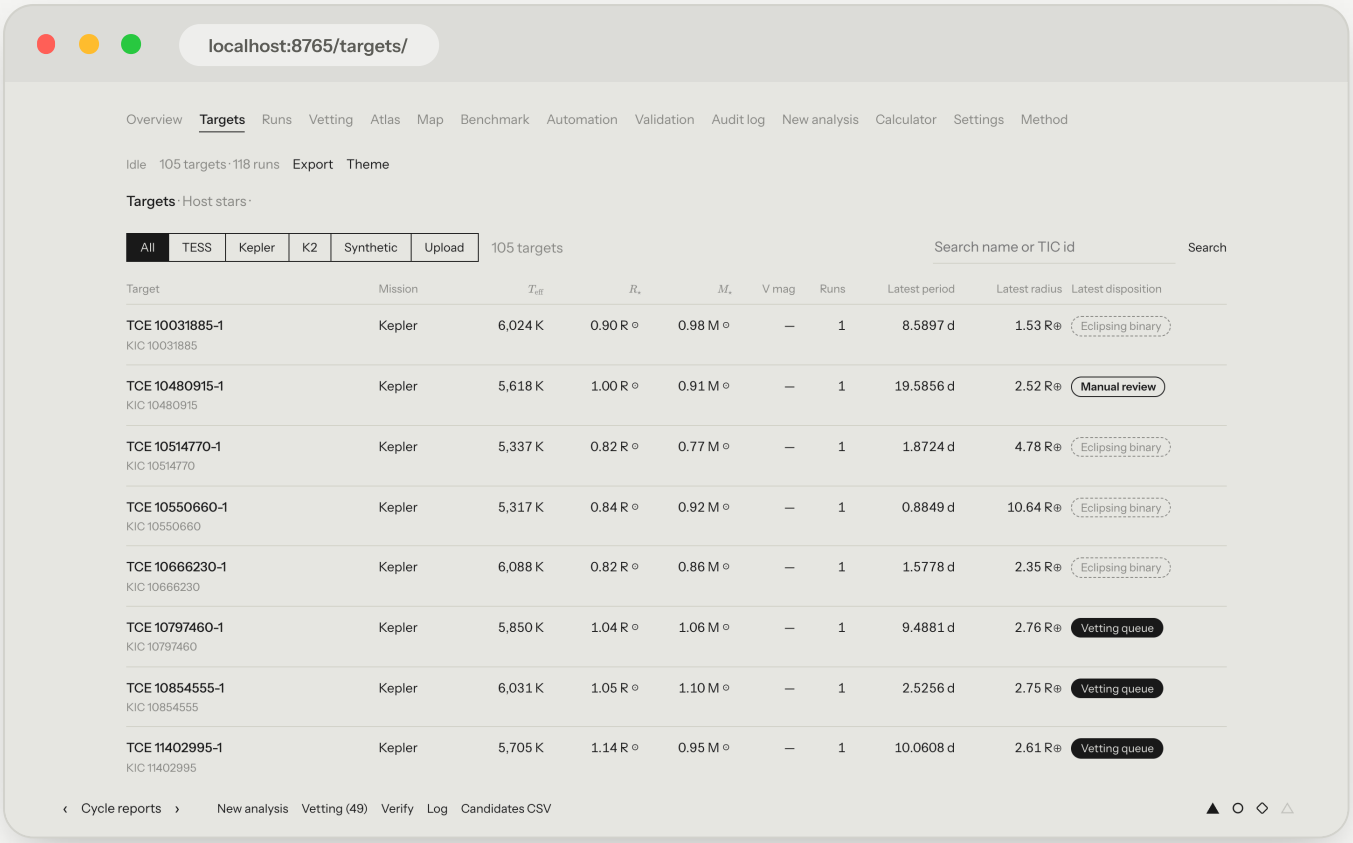
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RUN PAGE · AUDIT TRAIL

An audit trail for every stage

Every run closes with its audit trail: one schema-validated JSONL record per stage, each chained to the previous one with a SHA-256 hash.

- **Stages recorded.** Ingest, normalisation, outlier rejection, detrending, BLS search, phase folding, physics derivation, vetting diagnostics, CNN inference, refined analysis, archive cross-match and archive.
- **Verify integrity.** Recomputes the hash chain and reports whether any record has been altered.
- **Download JSONL.** The raw log for this run.
- **Reviewer decisions** and notes made on the vetting panel are written to the same trail.

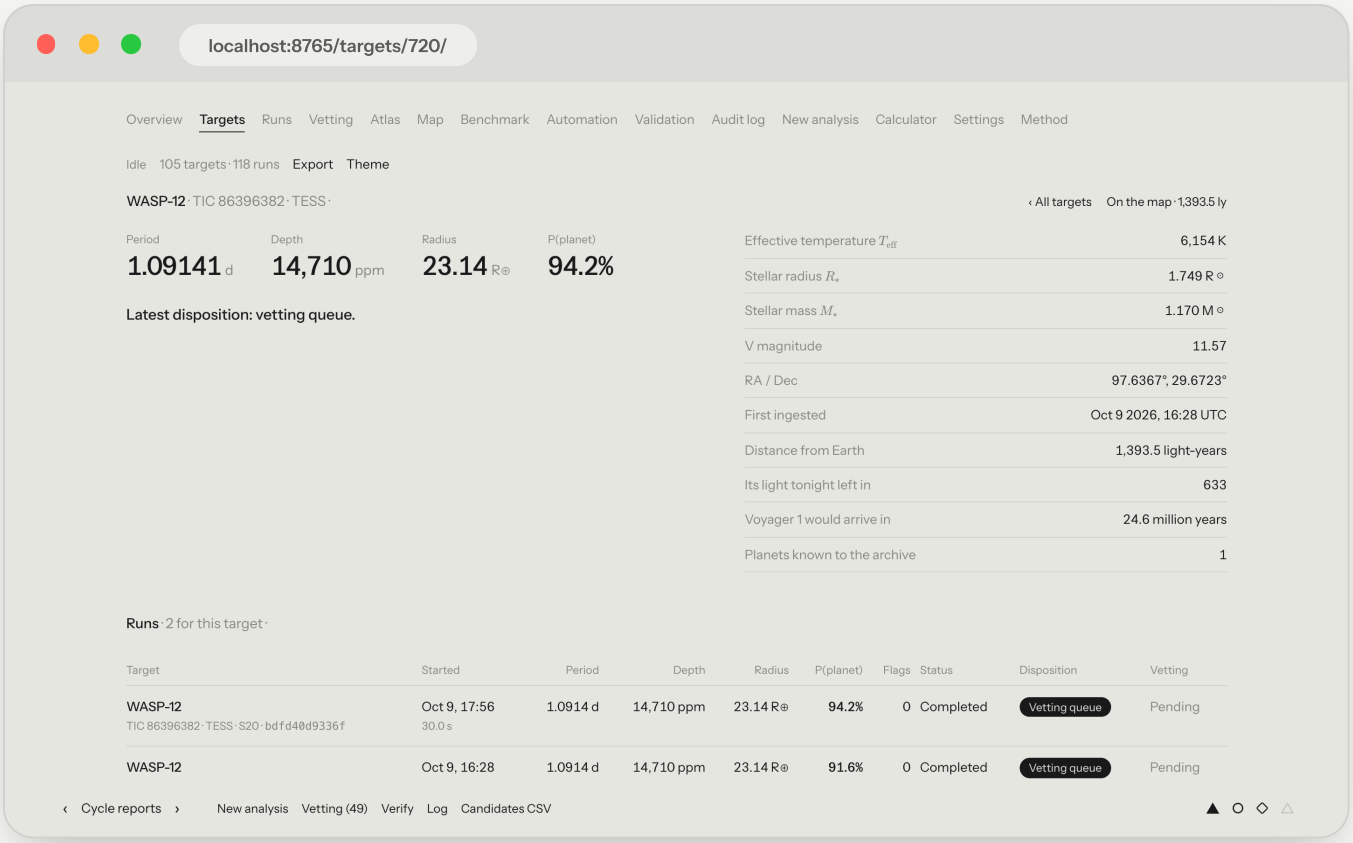


TARGETS · HOST STARS

Every host star in the archive

The targets page lists every host star in the archive, with its stellar parameters and the latest result found for it.

- **Filter by source.** All, TESS, Kepler, K2, Synthetic or Upload.
- **Search** by name or TIC id.
- **Columns.** Mission, effective temperature, stellar radius and mass, V magnitude, number of runs, latest period, latest radius and latest disposition.
- **Open a row** to see the star's own page.



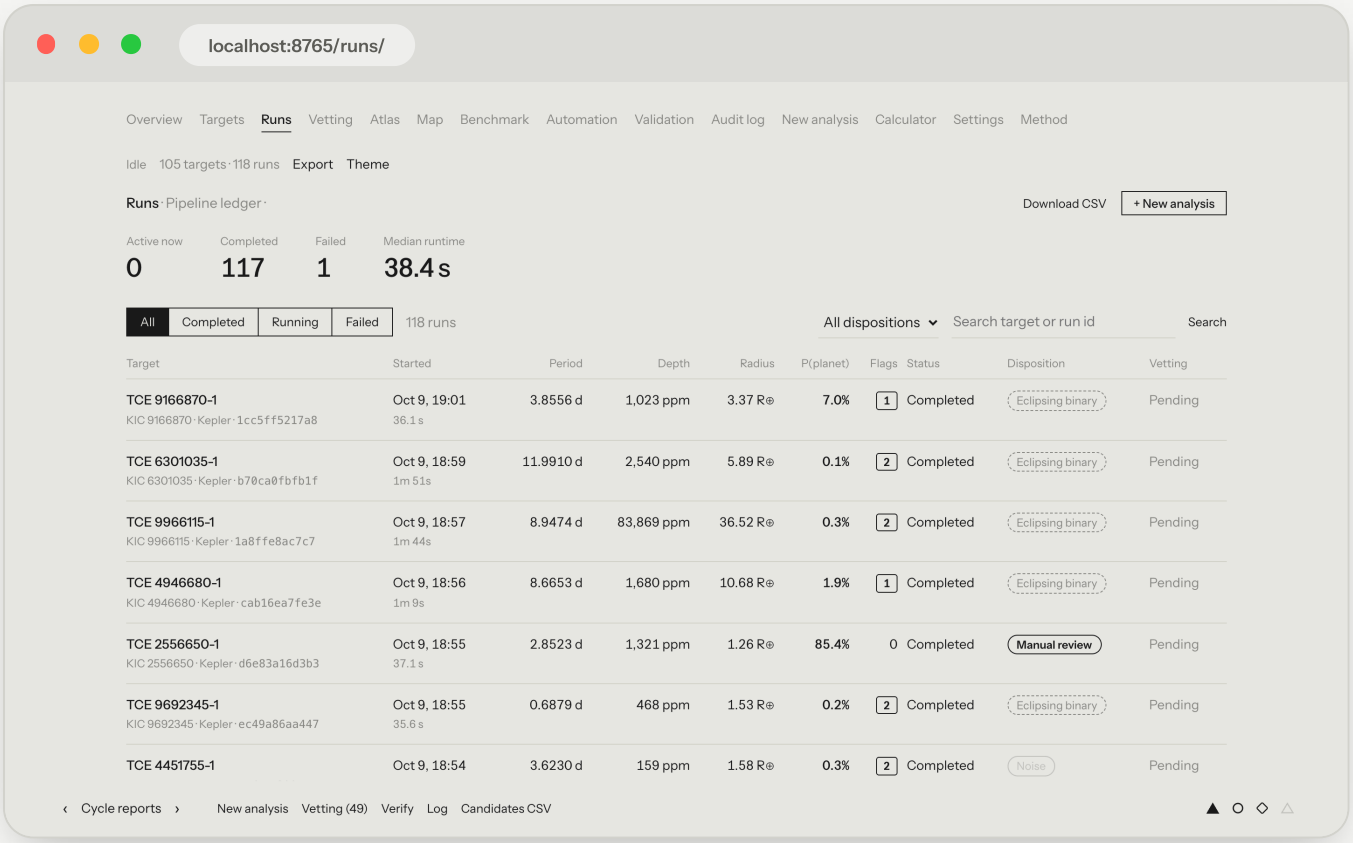
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TARGETS · ONE STAR

One star, in context

A target page puts one star in context: what the engine last measured on it, what the catalogue says about the star, and how far away it is in human terms.

- **Latest result.** Period, depth, radius, planet probability and disposition.
- **The star.** Temperature, radius, mass, V magnitude, sky position and when it was first ingested.
- **Distance.** WASP-12 is 1,393.5 light-years from Earth; the light seen tonight left in the year 633, and Voyager 1 would arrive in 24.6 million years.
- **Runs for this target.** Every analysis of this star, with a link to its place on the map.

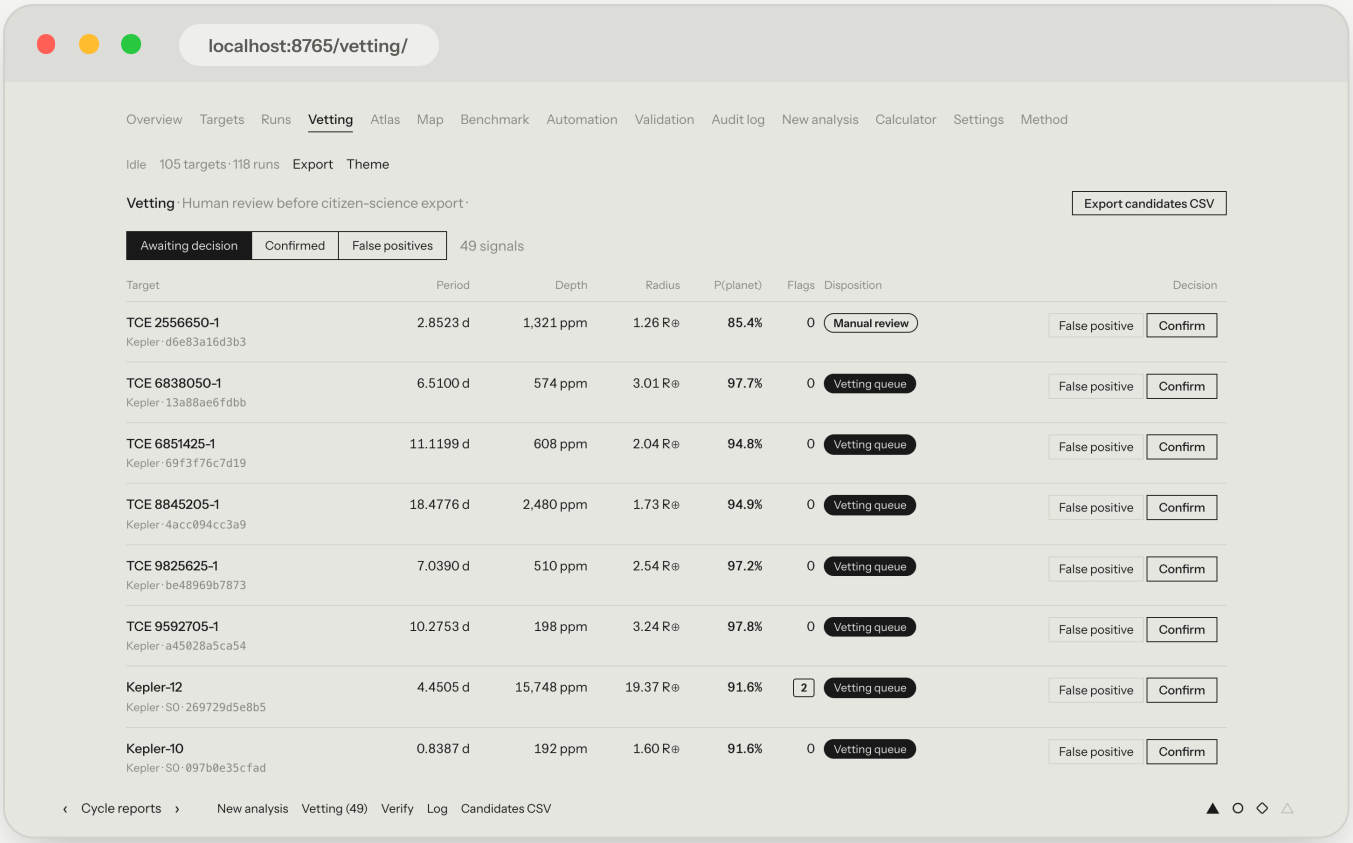


RUNS · PIPELINE LEDGER

The pipeline ledger

The runs page is the complete ledger of the pipeline's work: every run, its measurements, its status and its verdict.

- **Counters.** Active now, completed, failed, and median runtime.
- **Filter.** All, Completed, Running or Failed; narrow further by disposition, or search by target or run id.
- **Columns.** Period, depth, radius, planet probability, number of diagnostic flags, status, disposition and vetting decision.
- **Download CSV** exports the ledger; **+ New analysis** starts another run.

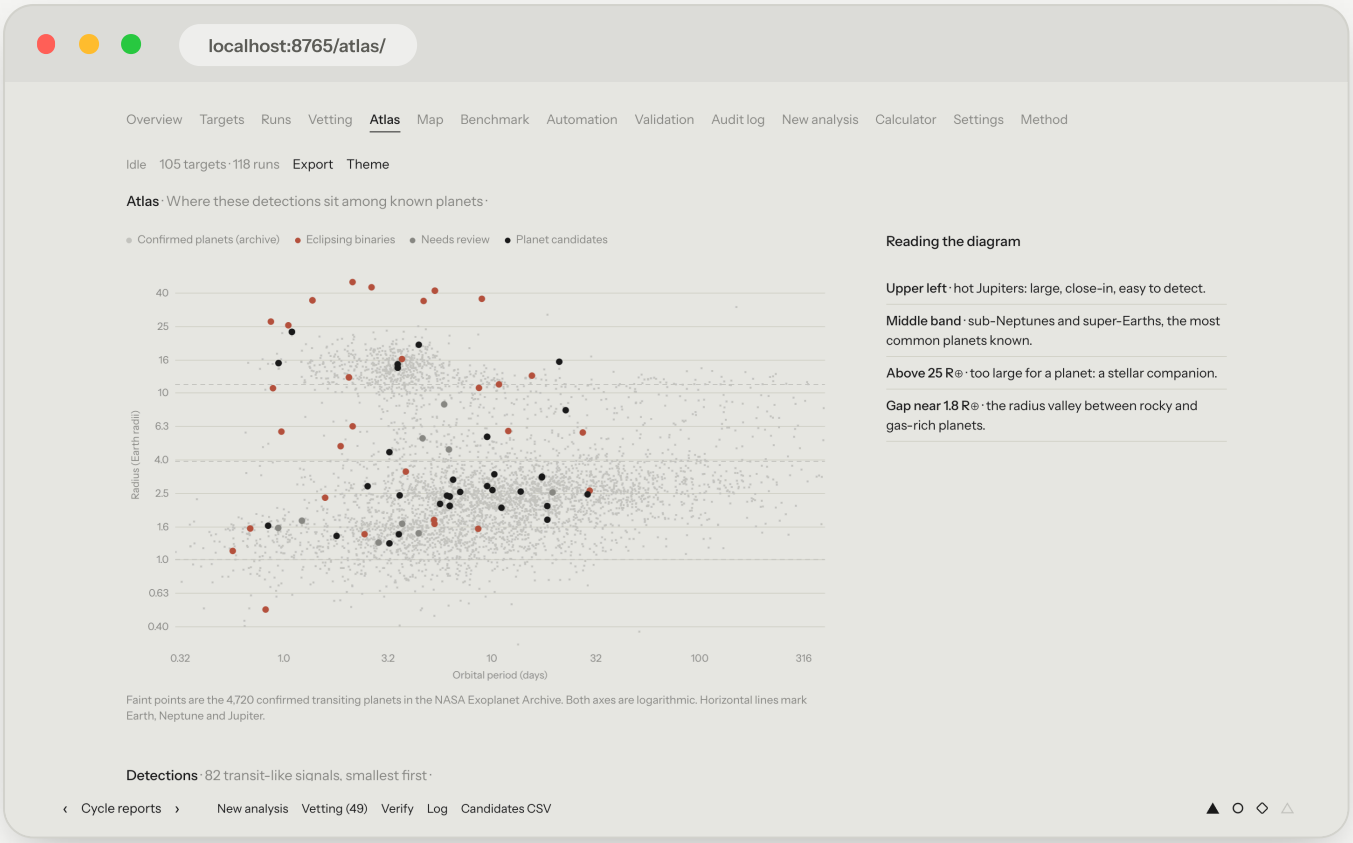


VETTING · HUMAN REVIEW

A human makes the final call

The engine never has the last word. Signals with a planet probability above the queue threshold and no high-severity flag arrive here for a person to decide.

- **Three tabs.** Awaiting decision, Confirmed and False positives.
- **Each row** shows the period, depth, radius, planet probability, flag count and disposition (Vetting queue or Manual review).
- **Confirm** or **False positive** records the decision against the run and in its audit trail.
- **Export candidates CSV** produces the list for citizen-science or follow-up use.
- The vetting-queue triangle beside every page shows the current queue size and links here.

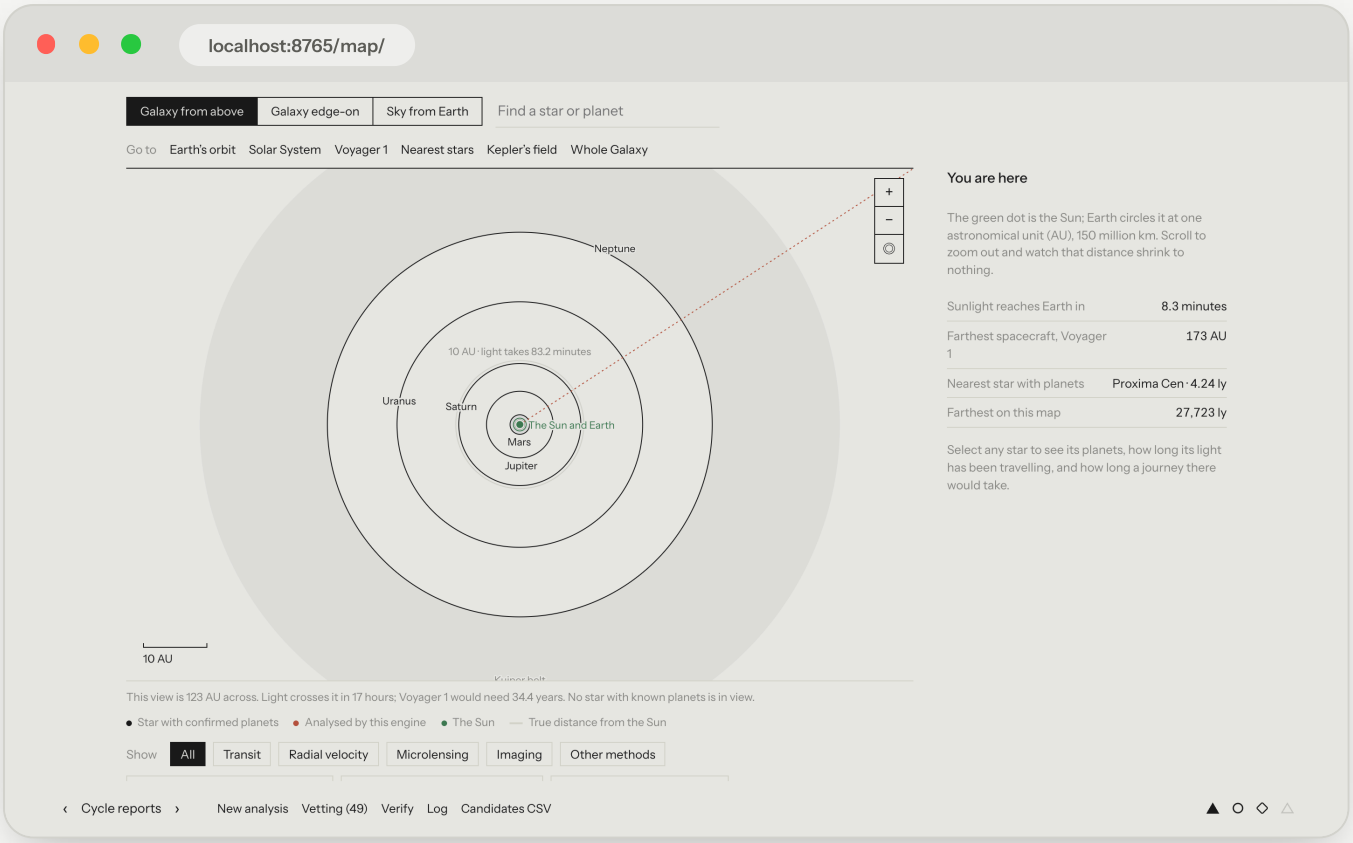


ATLAS · THE KNOWN POPULATION

Detections among the known planets

The atlas plots the engine's detections against the population of planets already known, by orbital period and radius, both on logarithmic axes.

- **Faint points.** The 4,720 confirmed transiting planets in the NASA Exoplanet Archive.
- **Engine results.** Planet candidates, eclipsing binaries and signals needing review are drawn on top.
- **Reading the diagram.** Upper left: hot Jupiters. Middle band: sub-Neptunes and super-Earths, the most common planets known. Above 25 R⊕: too large for a planet. The gap near 1.8 R⊕ is the radius valley between rocky and gas-rich planets.
- **Detections.** A table of the transit-like signals, smallest first, sits below the diagram.



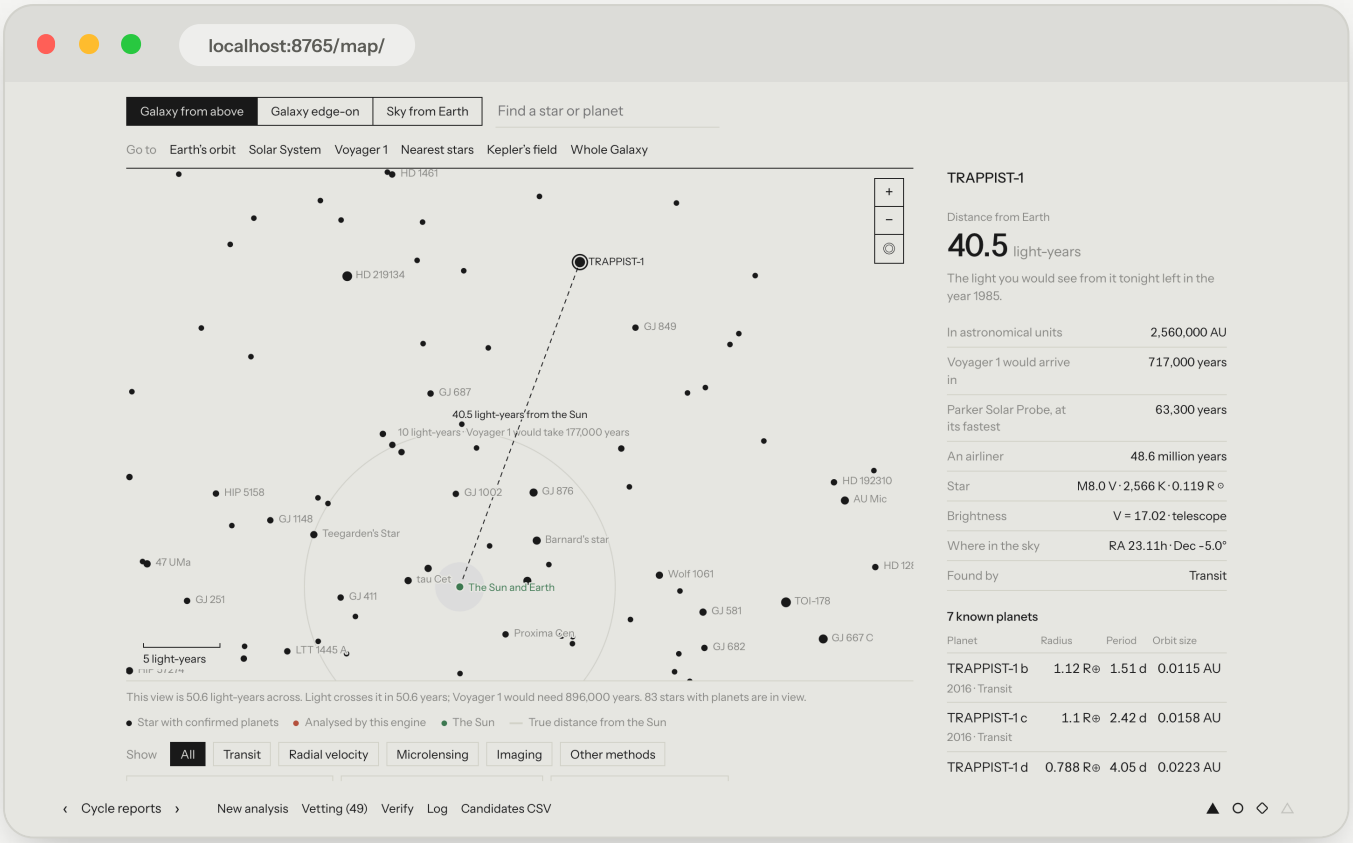
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MAP · YOU ARE HERE

The map begins at home

The map is a pannable, zoomable picture of real space. Every host star is placed from its measured sky position and distance in the NASA Exoplanet Archive; nothing is invented. It opens on the Solar System, to scale.

- **Three viewpoints.** Galaxy from above, Galaxy edge-on, and Sky from Earth.
- **Go to.** Earth's orbit, Solar System, Voyager 1, Nearest stars, Kepler's field and Whole Galaxy.
- **Find a star or planet** by name.
- **The readout** under the map says how wide the view is, how long light takes to cross it, and how long Voyager 1 would need.
- **Refresh from archive** updates the catalogue copy (6,417 planets in 4,814 systems here).



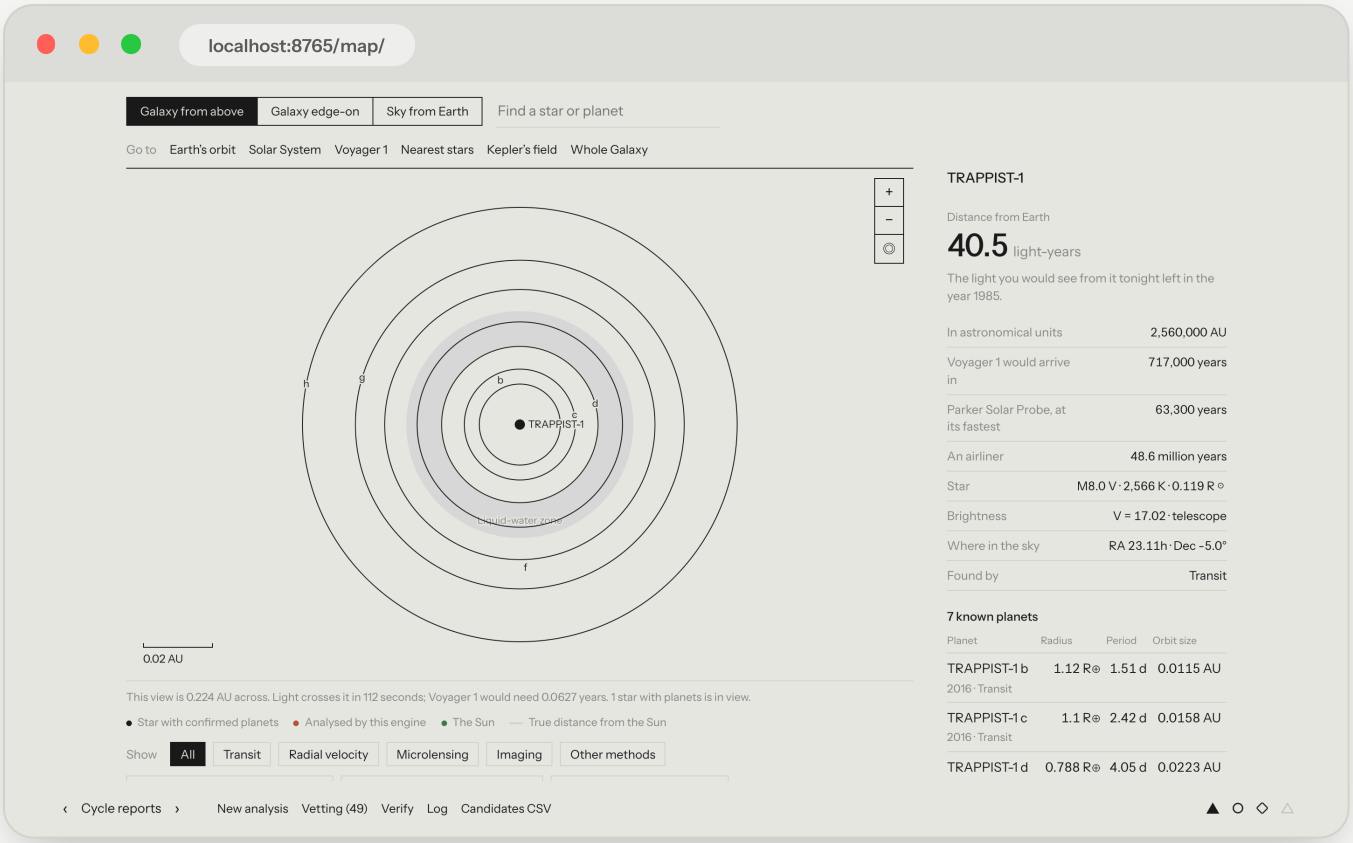
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MAP · THE NEAREST STARS

The neighbourhood, and how far it really is

Select any star and the panel explains its distance in terms a person can picture. TRAPPIST-1 is shown here among the stars within a few tens of light-years.

- **Distance from Earth.** 40.5 light-years; the light seen tonight left in 1985.
- **Travel times.** Voyager 1: 717,000 years. Parker Solar Probe at its fastest: 63,300 years. An airliner: 48.6 million years.
- **The star.** Spectral type, temperature, radius, brightness, sky position and discovery method.
- **Its planets.** All seven, with radius, period and orbit size.
- **Travel there from Earth** animates the whole way out.

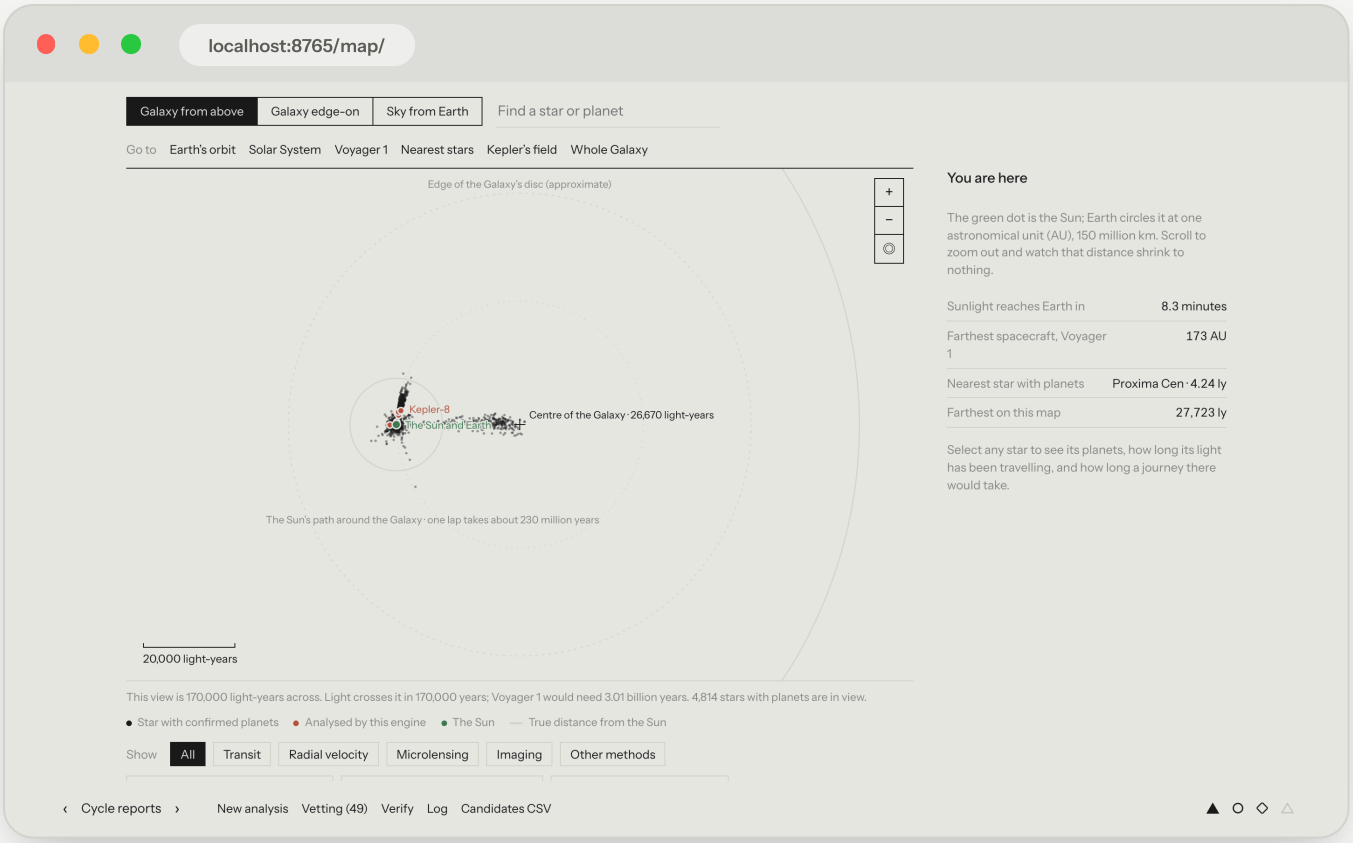


MAP · A SYSTEM UP CLOSE

Seven orbits and a liquid-water zone

Zoom all the way in and a star's orbits are drawn to scale with its liquid-water zone shaded. TRAPPIST-1's seven orbits would fit well inside Mercury's.

- **Scale.** This view is 0.224 AU across; light crosses it in 112 seconds.
- **Liquid-water zone.** The band where a planet receives enough starlight for liquid water.
- **Honest limits.** Orbits are drawn to scale in size but face-on, and planets are not placed along them, because orientation and present position are not in the catalogue.



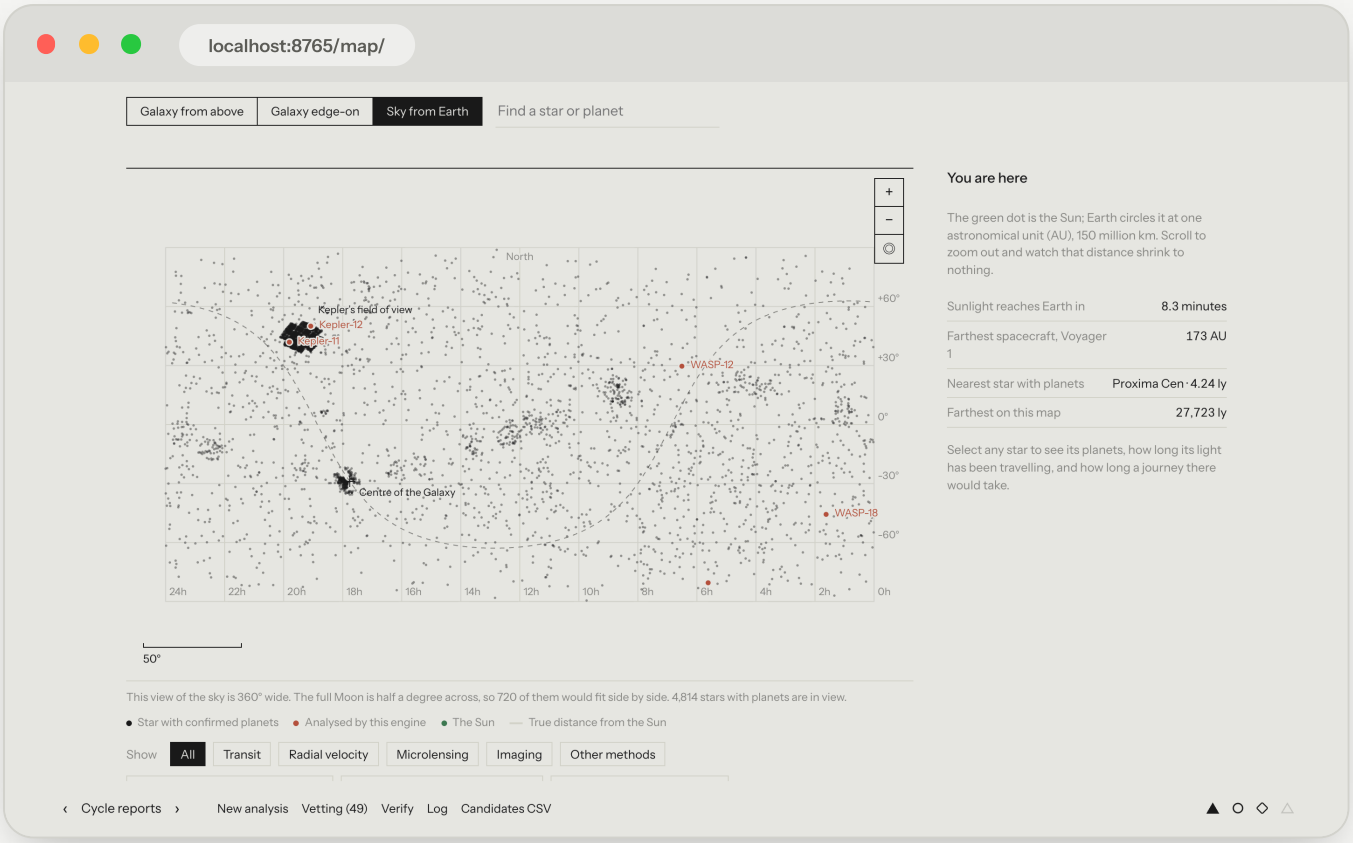
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MAP · THE WHOLE GALAXY

Every known planetary system

Zoomed out to 170,000 light-years, all 4,814 known planetary systems occupy a small patch around the Sun. The shape of that patch is the shape of where we have looked.

- **The spike** is the Kepler telescope's field of view.
- **The streak** towards the Galactic centre (26,670 light-years away) is microlensing discoveries.
- **The Sun's path** around the Galaxy is drawn too; one lap takes about 230 million years.
- Stars analysed by this engine are highlighted, and target and run pages link to their place here.

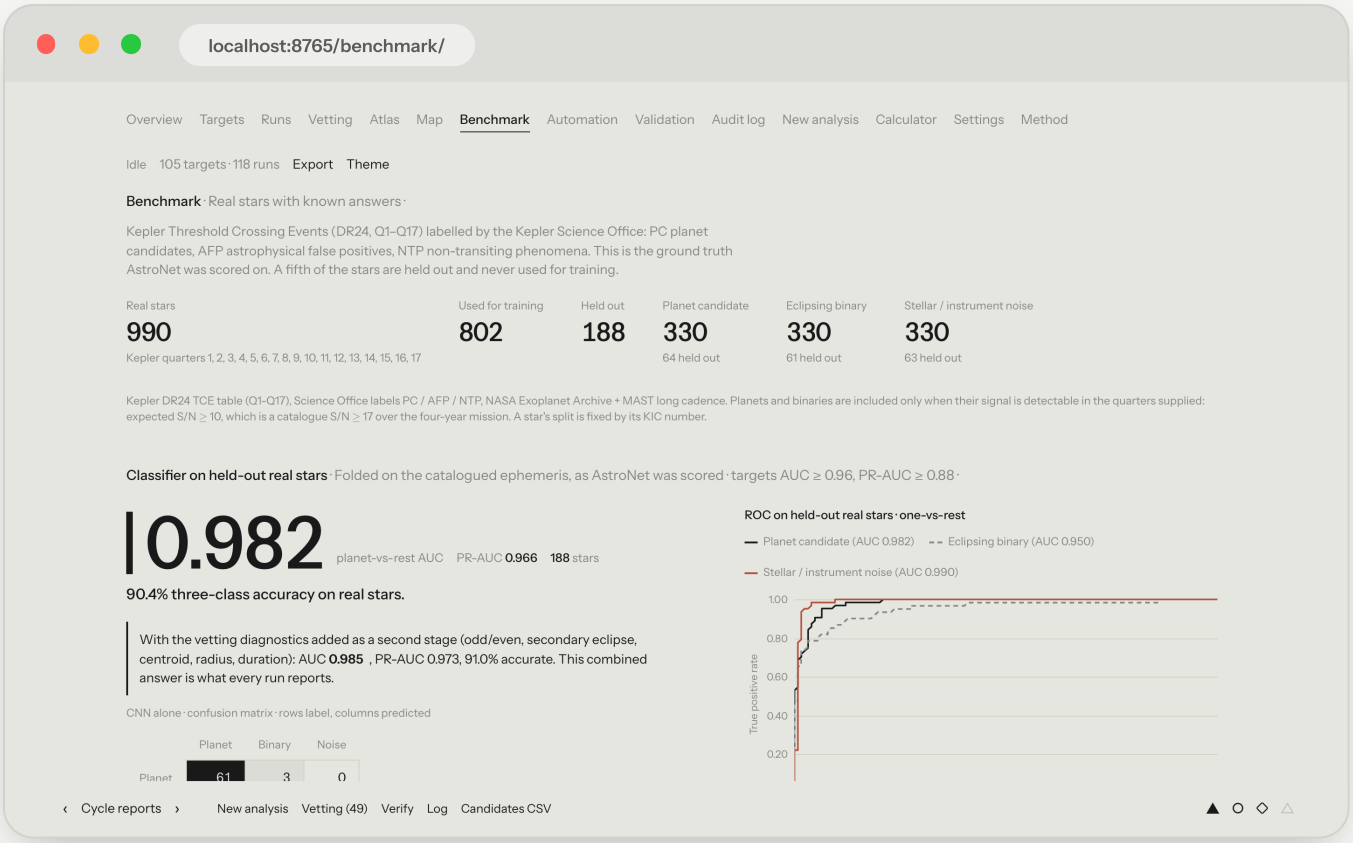


MAP · THE SKY FROM EARTH

The sky as seen from Earth

The third viewpoint lays the same systems out on the sky as seen from Earth, 360 degrees wide.

- **Kepler's field of view** appears as a dense patch; the Galactic centre is another.
- **Red points** are stars analysed by this engine.
- **Show filters.** All, Transit, Radial velocity, Microlensing, Imaging or Other methods; further filters select small planets, temperate planets and naked-eye stars.
- **The readout** compares the view with the width of the full Moon.

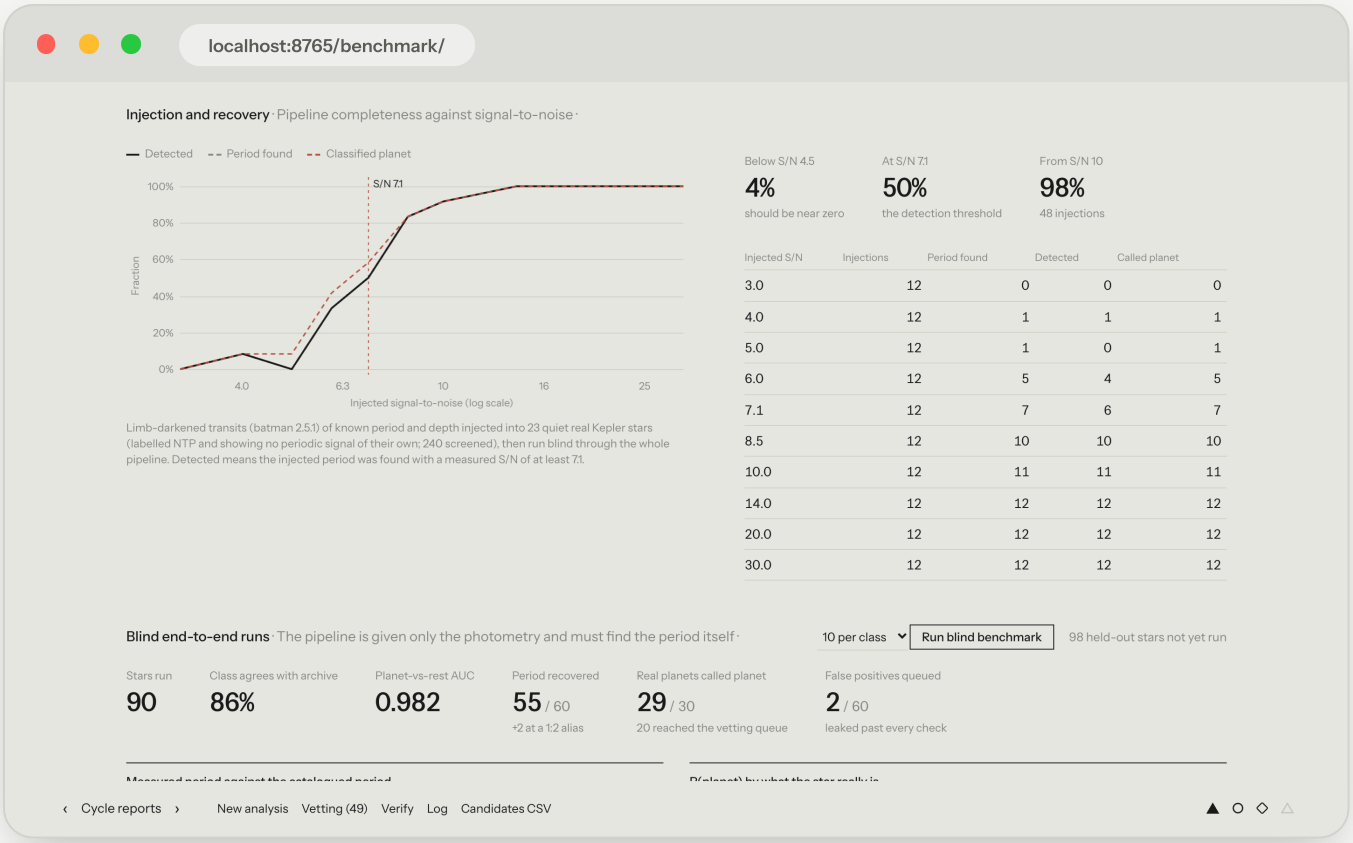


BENCHMARK · REAL STARS WITH KNOWN ANSWERS

Scored on real stars it never saw

The benchmark page scores the classifier on real Kepler stars labelled by the Kepler Science Office: planet candidates, astrophysical false positives and non-transiting phenomena. A fifth of the stars are held out and never used for training.

- **The data.** 990 real stars, 330 per class; 802 used for training, 188 held out. A star's split is fixed by its KIC number.
- **CNN ensemble alone.** Planet-versus-rest AUC 0.982, PR-AUC 0.966, 90.4% three-class accuracy.
- **With the second stage** (what every run reports): AUC 0.985, PR-AUC 0.973, 91.0% accurate.
- **Targets.** AUC at least 0.96 and PR-AUC at least 0.88.
- **ROC curves** and a confusion matrix show where the mistakes fall.

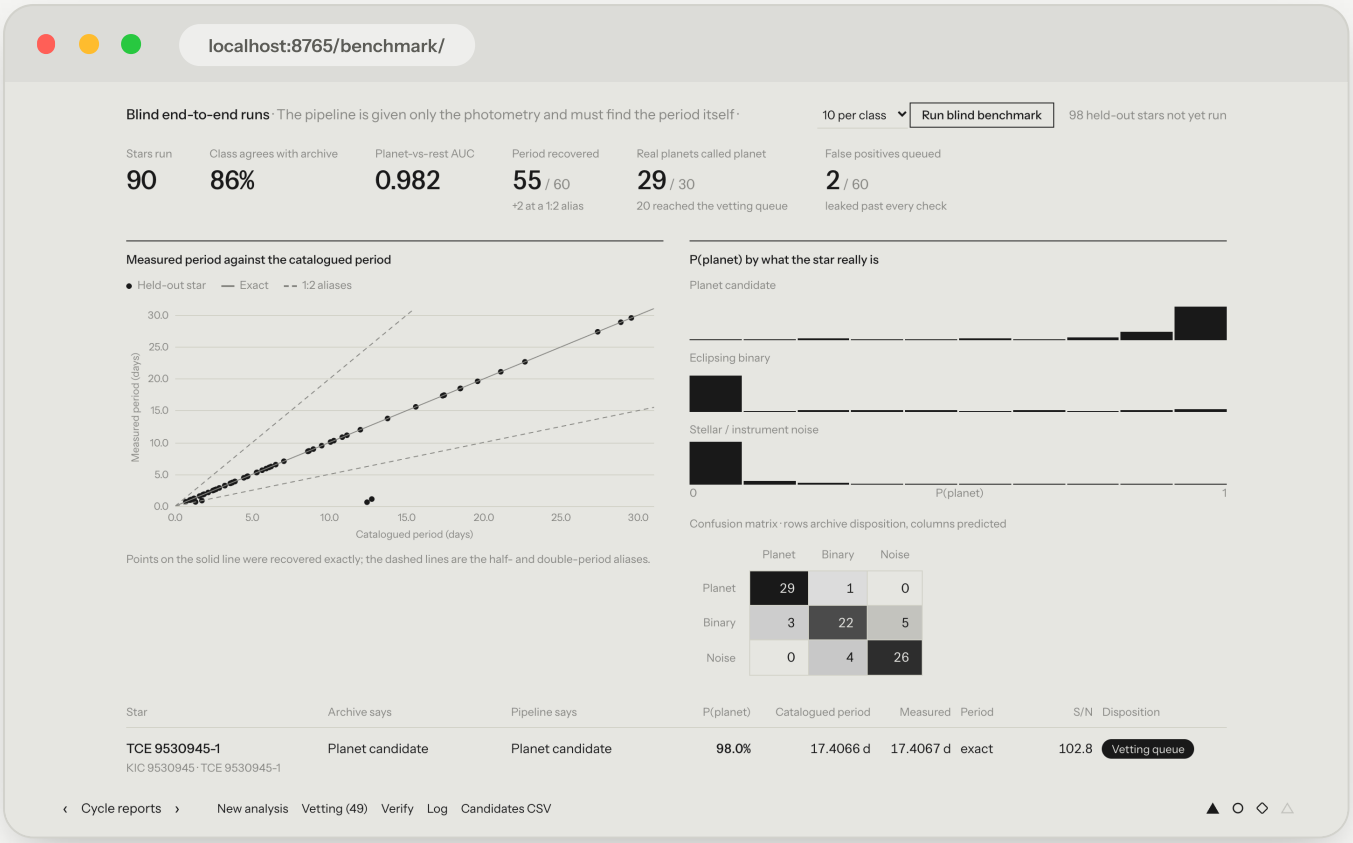


BENCHMARK · INJECTION AND RECOVERY

Injection and recovery

Injection and recovery measures completeness: limb-darkened transits of known period and depth are injected into quiet real Kepler stars and run blind through the whole pipeline.

- **From S/N 10:** 98% recovered.
- **At the 7.1 detection threshold:** 50%.
- **Below S/N 4.5:** 4%, which should be near zero and is the false-alarm floor.
- **The table** lists, for each injected S/N, how many periods were found, detected and called planet.
- On variable stars the further-signal search recovered 46 of 48 planets injected behind the star's own signal.

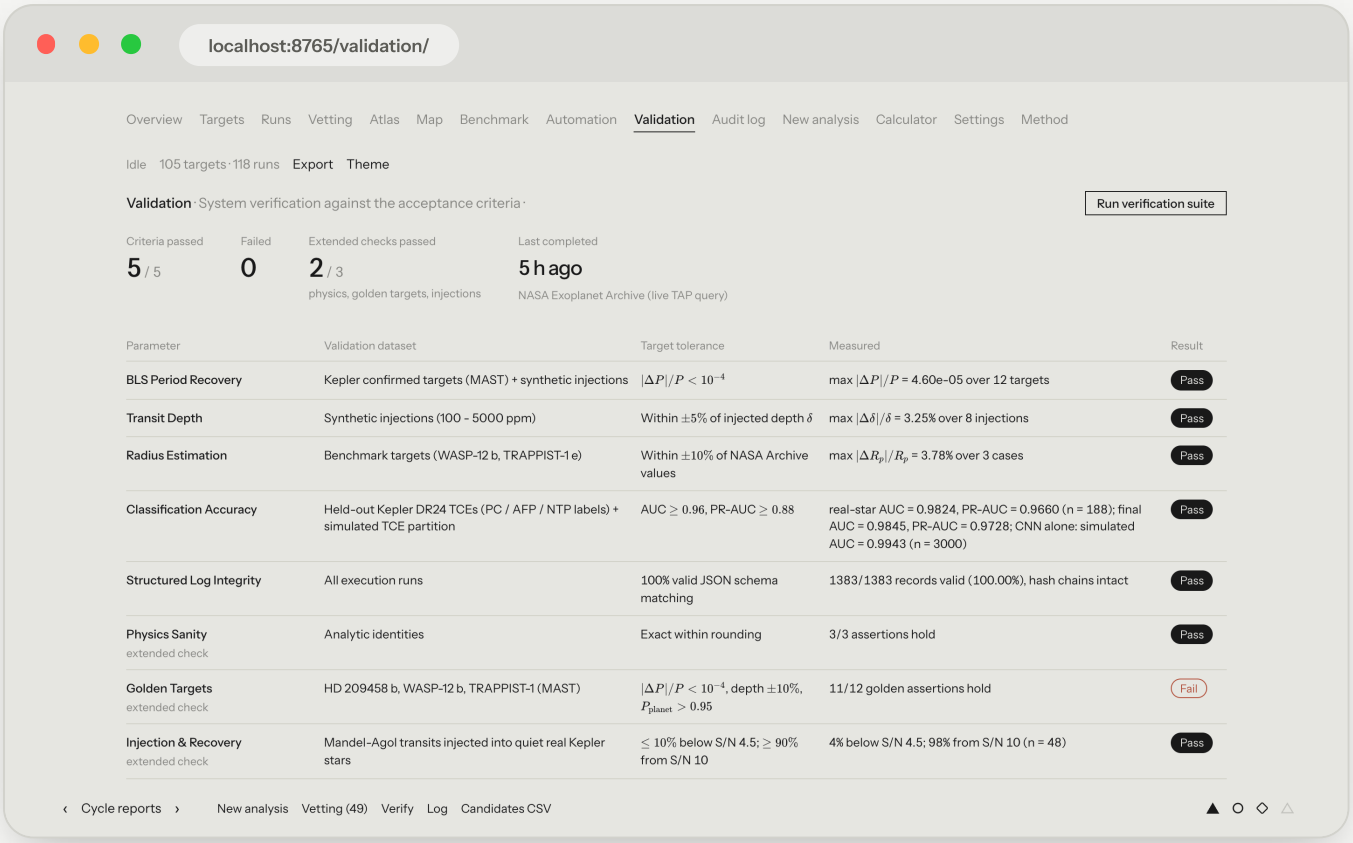


BENCHMARK · BLIND END-TO-END RUNS

Blind, end to end

In a blind run the pipeline is given only the photometry of held-out stars. It must find the period itself, derive the parameters and classify. Run blind benchmark queues more of the held-out stars.

- **Stars run:** 90. Class agrees with the archive label on 86%.
- **Period recovered:** 55 of 60 exactly, plus 2 at a 1:2 alias.
- **Real planets called planet:** 29 of 30, with 20 reaching the vetting queue.
- **False positives queued:** 2 of 60. One of them, TCE 6851425-1, carries an outdated label: NASA's current catalogue lists it as the confirmed planet Kepler-477 b.
- **Below.** Measured against catalogued period, the distribution of planet probability by true class, the confusion matrix, and every star with its result.



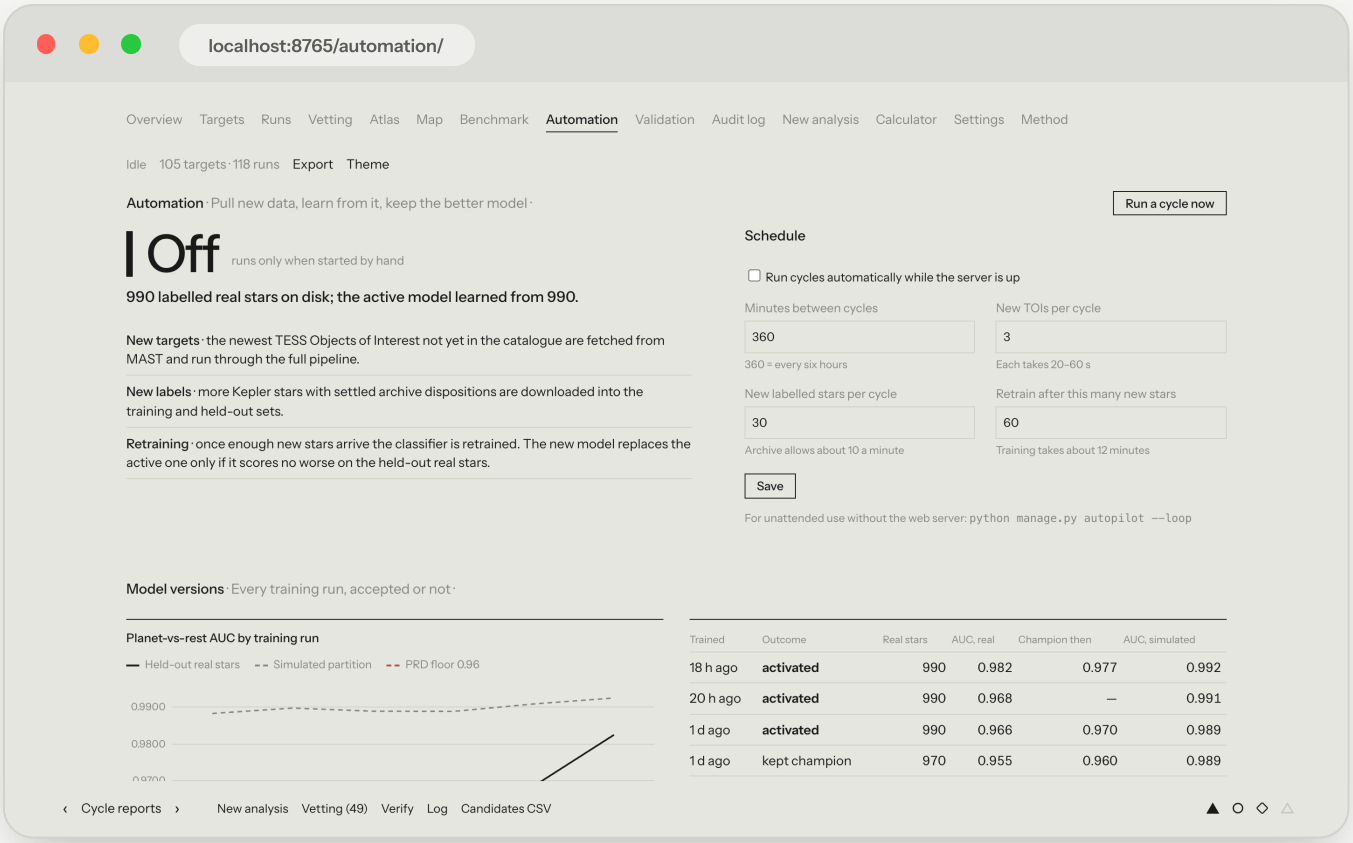
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VALIDATION · ACCEPTANCE CRITERIA

Checked against its acceptance criteria

Validation runs the verification suite against the system's written acceptance criteria and reports each result with the measured value beside the tolerance.

- **BLS period recovery.** Maximum error 4.6×10^{-5} against a limit of 10^{-4}.
- **Transit depth** within 3.25% (limit 5%); **radius** within 3.78% of NASA Exoplanet Archive values.
- **Classification accuracy** on held-out Kepler stars, and **structured log integrity**: every record valid, hash chains intact.
- **Extended checks.** Physics sanity, golden targets, and injection and recovery.
- **The one miss.** 11 of 12 golden-target assertions hold. WASP-12 b scores 0.942, under the 0.95 golden-target bar, though it still reaches the vetting queue.
- **Run verification suite** repeats the checks on demand.

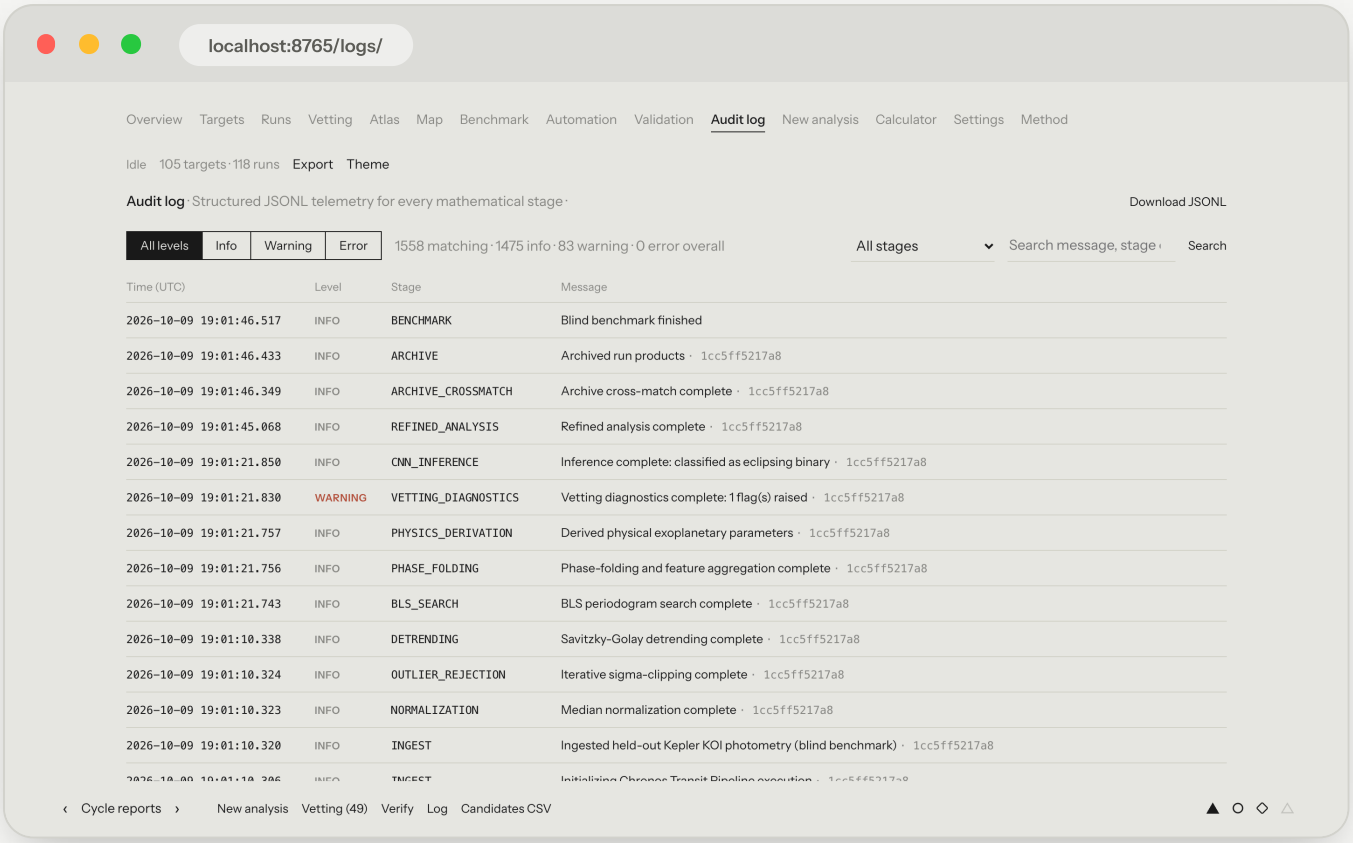


AUTOMATION · CHAMPION AND CHALLENGER

New data in, better model out

Automation runs cycles on a schedule so the engine keeps pulling new data and learning from it. It is off by default and can be started by hand with Run a cycle now.

- **New targets.** The newest TESS Objects of Interest not yet in the catalogue are fetched from MAST and run through the full pipeline.
- **New labels.** More Kepler stars with settled archive dispositions are downloaded into the training and held-out sets.
- **Retraining.** Once enough new stars arrive the classifier is retrained. The new model replaces the active one only if it scores no worse on the fixed held-out real stars.
- **Schedule.** Minutes between cycles, new TOIs per cycle, new labelled stars per cycle, and how many new stars trigger a retrain.
- **Model versions.** Every training run, accepted or not, with its AUC on real and simulated data.
- For unattended use without the web server: `python manage.py autopilot --loop`.

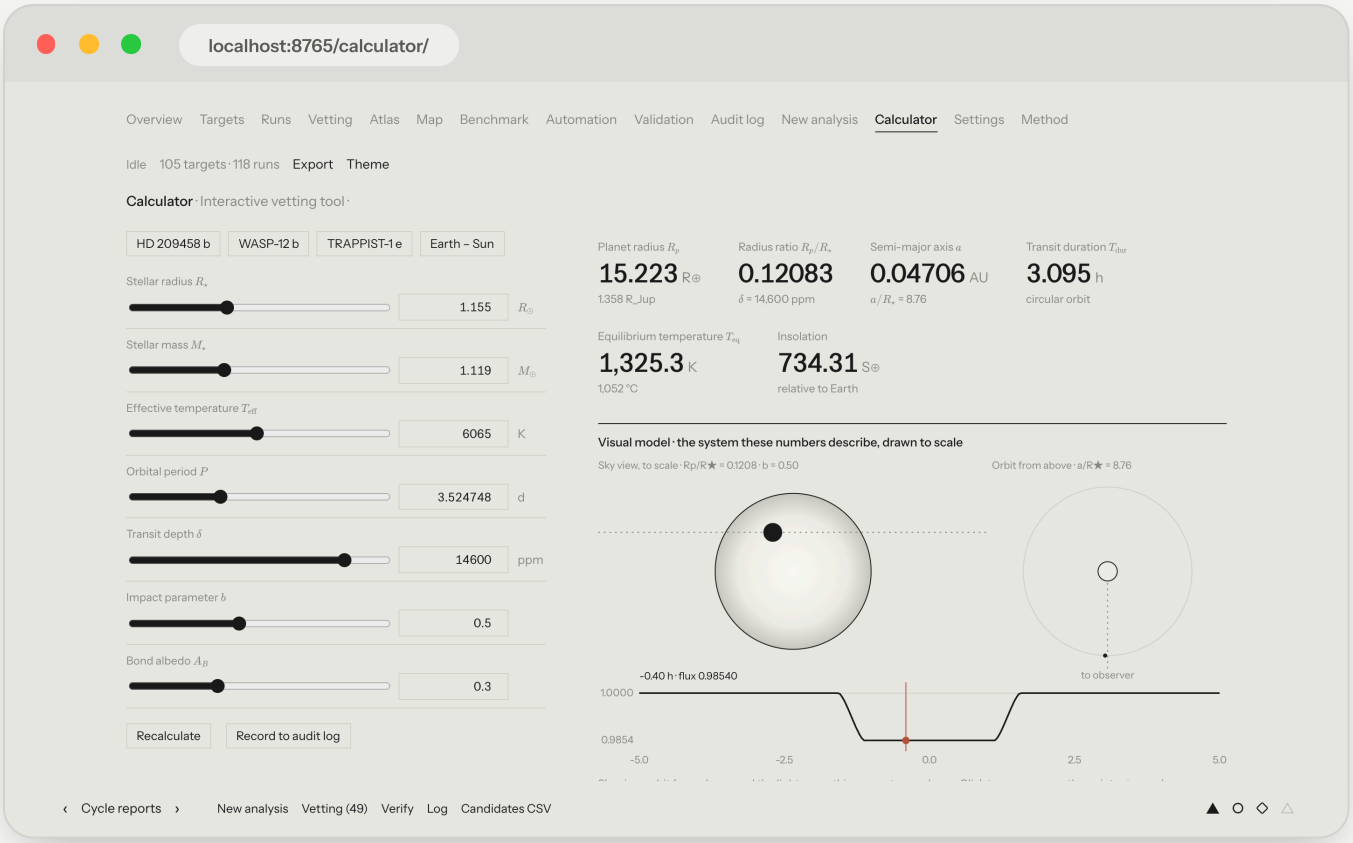


AUDIT LOG · ALL RUNS

Every stage of every run

The audit log gathers the structured telemetry from every mathematical stage of every run into one searchable table.

- **Filter by level.** All, Info, Warning or Error, with running totals.
- **Filter by stage** or search the message text.
- **Each record** carries its UTC timestamp, level, stage, message and run id.
- **Download JSONL** exports the complete log.

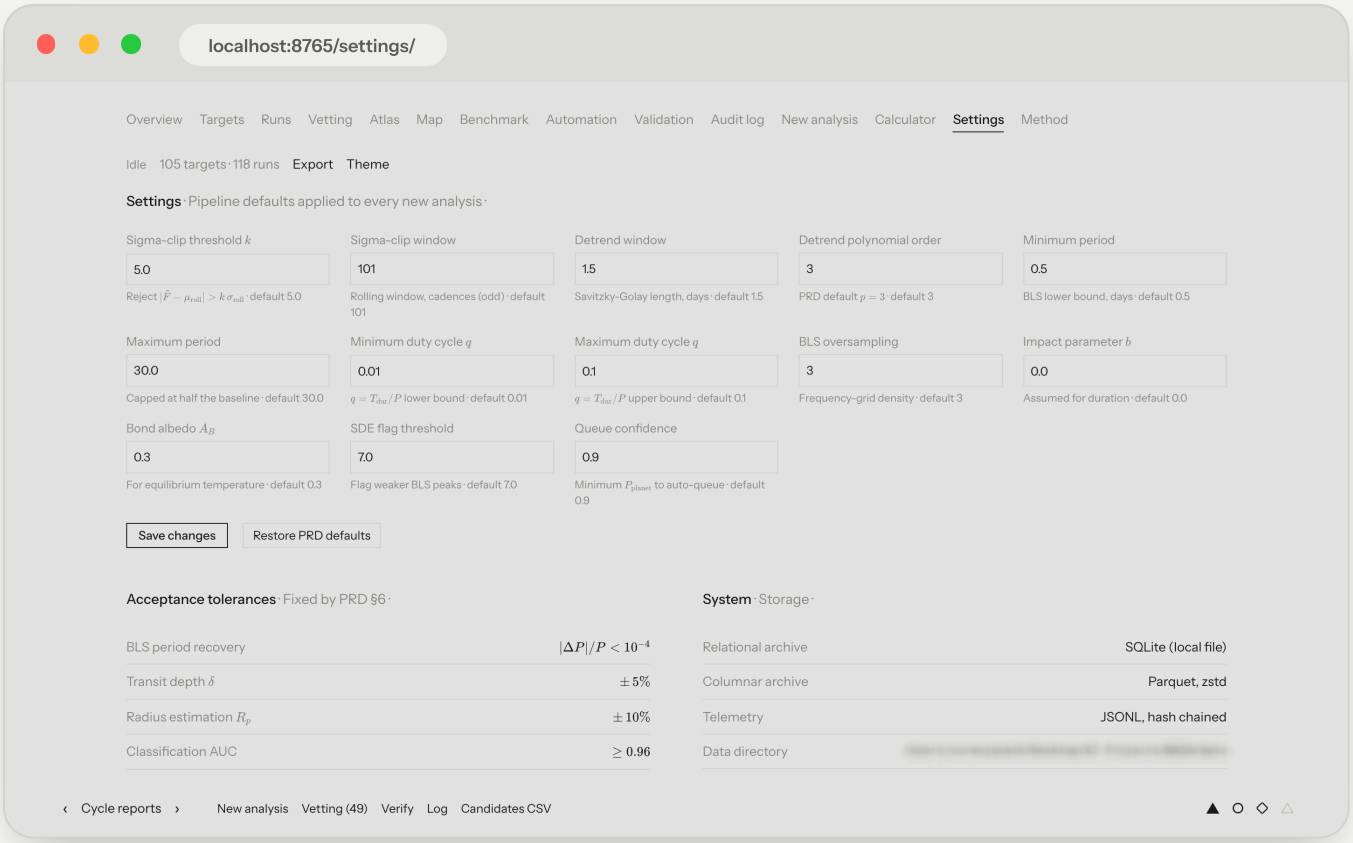


CALCULATOR · INTERACTIVE VETTING TOOL

Work the physics by hand

The calculator applies the engine's physics to numbers you choose, with the result drawn to scale as you change them.

- **Presets.** HD 209458 b, WASP-12 b, TRAPPIST-1 e and Earth–Sun.
- **Inputs.** Stellar radius, mass and temperature, orbital period, transit depth, impact parameter and Bond albedo.
- **Outputs.** Planet radius, radius ratio, semi-major axis, transit duration, equilibrium temperature and insolation relative to Earth.
- **Visual model.** The same sky view, orbit and light curve used on run pages.
- **Record to audit log** saves a calculation alongside the pipeline's own records.



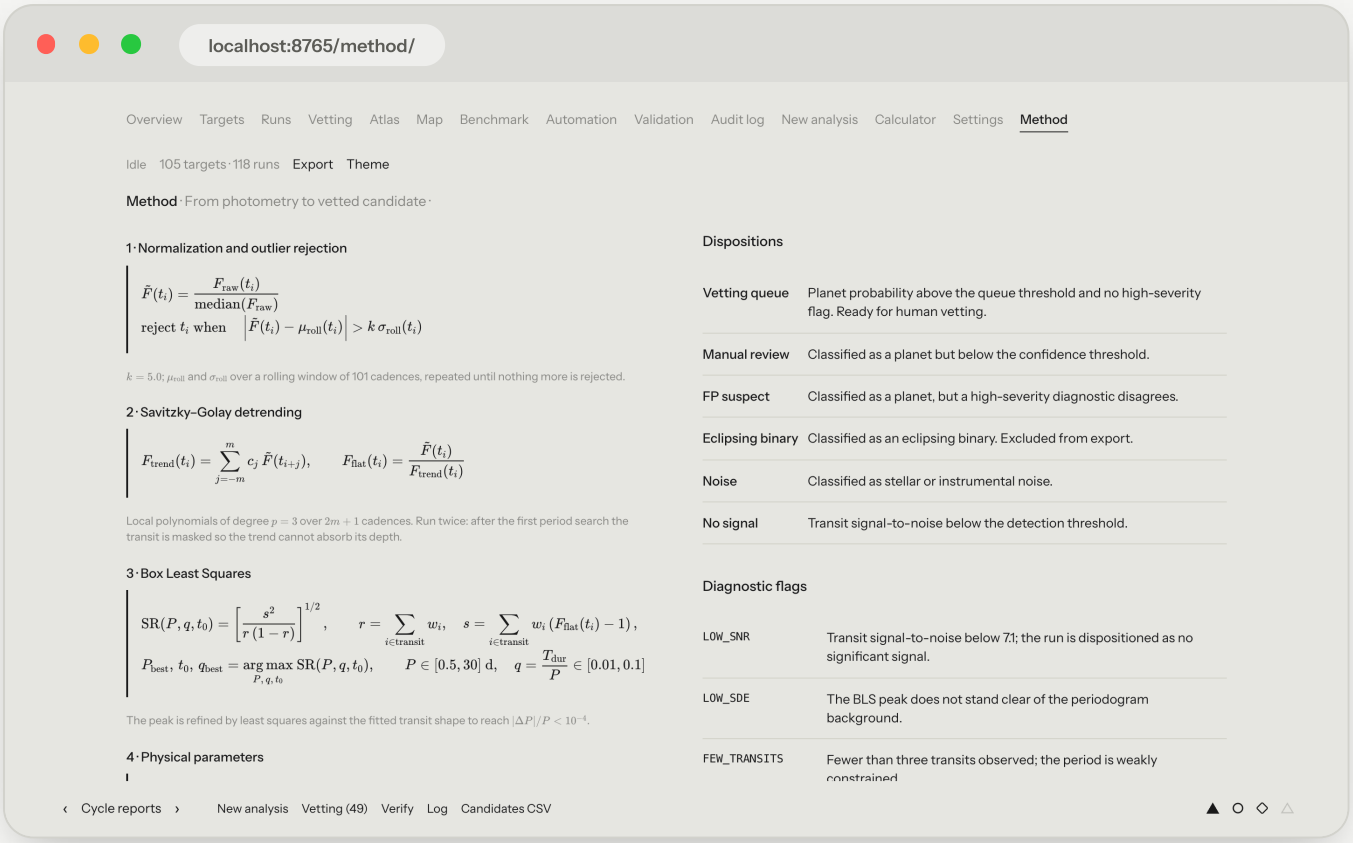
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SETTINGS · PIPELINE DEFAULTS

Pipeline defaults in one place

Settings holds the defaults applied to every new analysis. Each field shows its meaning and its original default; Restore PRD defaults puts everything back.

- **Cleaning.** Sigma-clip threshold and window, detrend window and polynomial order.
- **Search.** Minimum and maximum period, minimum and maximum duty cycle, and BLS oversampling.
- **Physics and decisions.** Impact parameter, Bond albedo, the SDE flag threshold, and the queue confidence a signal needs to be auto-queued for vetting (0.9).
- **Acceptance tolerances** are fixed and shown read-only.
- **System.** The relational archive (SQLite or PostgreSQL), the columnar archive (Parquet), the telemetry format and the data directory.



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METHOD · FROM PHOTOMETRY TO VETTED CANDIDATE

The mathematics, written out

The method page documents the pipeline in full, with the formula for each stage typeset and the parameters used.

- **Stages.** Normalisation and outlier rejection, Savitzky-Golay detrending, Box Least Squares, and the physical parameters.
- **Dispositions.** Vetting queue, Manual review, FP suspect, Eclipsing binary, Noise and No signal, each defined.
- **Diagnostic flags.** Every flag the engine can raise, such as LOW_SNR, LOW_SDE and FEW_TRANSITS, with what triggers it.
- **Plots.** How to read each plot that appears on a run page.

IN SUMMARY

From starlight to a vetted candidate

Chronos Transit Engine takes raw starlight to a measured planet, a decision you can audit, and a place on the map. It is candid about what it cannot yet do.

- **No confirmed new planet.** Across the real stars analysed, every signal sent to the vetting queue is a NASA-confirmed planet except KOI-143.01. Confirming a planet takes independent follow-up.
- **Variable stars.** Two of 48 injected planets are still missed, and 2 of 45 variable stars get a spurious extra signal. Additional signals are candidates for inspection.
- **Speed.** A four-year Kepler star takes 40 to 190 seconds end to end; a 27-day TESS sector takes a few seconds.
- **The map.** Orbits are drawn face-on, and 28 planets with no measured distance are left off.
- **Useful commands.** `manage.py seed`, `build_kepler_dataset`, `train_model`, `benchmark`, `injection_study`, `autopilot`, `verify`.

Thank you

This guide covered every page of Chronos Transit Engine: starting an analysis, reading a run from verdict to audit trail, human vetting, the atlas and the map, the benchmarks that score the system on real stars, and the automation, settings and method behind it.

FINDS PLANETS IN STARLIGHT